

# Alpha Asymmetry in Foreign Exchange Markets

## *An Investigation of Exploitability*

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### Abstract

Skewness-based trading rules presuppose asymmetrically distributed signals. We test that premise, and whether a rule built on it earns money, for five technical transformations of the EUR/JPY price (“alpha signals”), using weekly indicative Yahoo Finance mid-quotes from January 2016 to August 2025 (504 weeks). Under block-bootstrap inference only the volatility-expansion signal is robustly skewed (1.75, 95% CI [1.18, 2.16]). The tail signal’s skewness changes sign across three defensible weekly aggregations of one daily rule, and the positive tail skewness reported in an earlier version described an unsigned magnitude series, right-skewed by construction.

A skewness-threshold rule executed at the first trading-session open after each Friday signal returns −0.73% cumulative gross over 15 holding episodes, with a Sharpe ratio indistinguishable from zero. No break-even cost exists. The result is not stable across specifications: all four execution timings lose money, and four pre-specified readings of the entry rule span 9.04 percentage points and change sign. Walk-forward selection leaves the rule nearly dormant, and proxy factor regressions show no positive residual alpha (intercept  $t = -0.45$  on in-position weeks). Neither White’s Reality Check nor Hansen’s SPA rejects the null that no candidate rule beats a zero-return benchmark. Earlier versions, now superseded, misapplied the holding rule and priced at the Friday close. The evidence covers one currency pair and does not model capacity, financing or per-order costs. Within that scope the evidence does not establish an exploitable edge, and the results depend on construction choices the original specification did not justify.

**Keywords:** null result, alpha asymmetry, foreign exchange, skewness, market efficiency, extreme value theory

**JEL Codes:** G11, G14, G15, C58

## 1 Introduction

Asset returns are not Gaussian: they have fat tails (Mandelbrot, 1963; Cont, 2001) and asymmetric distributions (Harvey and Siddique, 2000). Separately, simple rules that trade on past prices, from cross-sectional momentum in equities (Jegadeesh and Titman, 1993) to time-series momentum across asset classes (Moskowitz et al., 2012), have long been used to probe the efficient market hypothesis (Fama, 1970). This paper joins the two concerns for one liquid currency pair. If the signals a simple rule reads are asymmetrically distributed, can a rule that conditions on that asymmetry earn money under realistic implementation choices? The signals are five technical transformations of the EUR/JPY price series: the signed size of large daily moves, a normalized five-day return, a deviation from a sixty-day moving average, a ratio of realized volatilities, and a fixed-rate correlation proxy (Section 2.2). We call them alpha signals for continuity with the earlier version of this paper; they are not estimates of risk-adjusted excess return.

The preference for skewed returns has deep roots in asset pricing theory. [Kraus and Litzenberger \(1976\)](#) showed that investors prefer positive skewness and dislike negative skewness, implying that assets with lottery-like payoffs command lower expected returns. Subsequent work has confirmed that skewness affects both individual asset pricing ([Harvey and Siddique, 2000](#); [Bali et al., 2011](#)) and portfolio construction ([Mitton and Vorkink, 2007](#); [Brunnermeier et al., 2007](#)).

Our test has two links. The first is statistical: does a signal’s skewness survive inference that allows for the serial dependence its rolling construction creates? The second is economic: does a rule that enters when a signal’s rolling skewness exceeds a threshold earn a positive return that survives costs, training-only parameter selection and a data-snooping correction? A positive result would require both links to hold; a failure at either is a null, and we report where in the chain it occurs. The proposition at stake is whether distributional asymmetry in simple price-based signals is a usable source of return in a liquid currency pair. The evidence comes from one pair, weekly indicative mid-quotes and a rule that holds a position in 55 of 503 weeks, so it does not reach other pairs, intraday horizons, executable quotes, capacity or financing. We address four questions:

1. Do alpha signals in forex markets deviate significantly from normal distributions? *(Mostly: four of five types reject normality; normality is not rejected for the normalized-return signal)*
2. Do these deviations manifest as exploitable skewness and heavy tails? *(Largely no: one signal is robustly skewed, the tail signal skews negative, but the estimate is fragile, and the tail-shape estimate is too imprecise to establish Pareto-type heavy tails)*
3. Do distributional asymmetries persist across market regimes? *(No: the tail signal’s skewness flips sign across regimes)*
4. Do asymmetry-aware strategies outperform benchmarks? *(No: the baseline returns  $-0.73\%$  gross, with a Sharpe ratio indistinguishable from zero, and its sign changes across pre-specified entry rules; it is dormant under walk-forward selection, and no candidate significantly beats the zero-return benchmark)*

The idea fails, and it fails first at the statistical link: an unsigned-magnitude construction in the earlier version had manufactured an appearance of pervasive positive skewness that corrected measurement does not support. The economic link fails as well. The one robust asymmetry, in the volatility-expansion signal, is not shown to predict the direction of returns, and the rule, which reads the skewness of two signals whose full-sample intervals include zero, has no stable economic result. Null results of this kind are underreported in quantitative finance ([Harvey, 2017](#)), yet they prevent wasted research effort and capital allocation to spurious patterns.

The closest work falls into four groups. On technical trading in currencies, [Qi and Wu \(2006\)](#) apply White’s Reality Check to 2,127 rules and find that the best remain significantly profitable over their full sample, with profitability much weaker in the more recent period. [Neely et al. \(2009\)](#) show in true out-of-sample tests that the excess returns of filter and moving-average rules had disappeared by the early 1990s, a pattern consistent with the adaptive markets hypothesis and not with efficient markets. [Hsu et al. \(2016\)](#), testing over 21,000 rules on 30 currencies with a stepwise correction for data snooping, still find substantial predictability, varying across subperiods and currency groups. Against that evidence a twelve-rule universe on one pair says little about technical trading in general; what this paper adds is one asymmetry-conditioned rule and the specification sensitivity reported in Sections 4.4 and 4.5. On skewness as a currency signal, [Jiang et al. \(2019\)](#) find a positive and significant relation between return

skewness and future currency excess returns, and [Li et al. \(2026\)](#) find that a skewness risk premium measured from semivariances and currency options earns high long-short returns and is priced in a cross-section of 60 currency portfolios. Both study the skewness of currency returns across many currencies; we study the skewness of five price-based signals on a single pair. On currency factors and tail risk, [Fan et al. \(2025\)](#) dynamically optimize currency factor strategies, including carry, momentum, and value, via spot and forward trading, evaluating 24,336 portfolio optimization approaches and finding that optimized factors significantly outperform naïve constructions after correcting for data snooping bias. [Hertrich \(2025\)](#) conditions G10 currency strategies on forward-looking Conditional Value-at-Risk and notes that the carry trade risk premium has remained unexpectedly low since the global financial crisis despite its negative covariance with global FX volatility, which that study reads as a failure of currency pricing theory. Fan et al. optimize how factors are constructed across a currency cross-section, and Hertrich conditions a G10 cross-section on forward-looking tail risk; we condition a single pair’s rule on the realized skewness of its own price-based signals. On measurement, [Bai and Ng \(2005\)](#) derive skewness and normality tests that remain valid under serial correlation. We use a block bootstrap for the same purpose in our skewness intervals (Section 3.2), but the normality tests in Table 3 assume independent observations and do not apply their correction.

Our empirical contribution is threefold, and we state it narrowly. First, we characterize the distributional properties of five alpha signals in EUR/JPY data (Sections 3.1 and 3.2); one of them, the volatility-expansion signal, survives dependence-robust inference under the paper’s primary construction. Second, we report a corrected analysis of the trading rule (Sections 3.3 and 4): it returns  $-0.73\%$  gross with a Sharpe ratio indistinguishable from zero, walk-forward selection leaves it nearly dormant, and data-snooping-corrected tests do not reject the null of no superior performance against a zero-return benchmark across a twelve-strategy universe. Third, we document the sensitivity of those results to three choices the original specification did not argue for. Their statistical behavior differs, so we report them separately. Under the weekly aggregation of the tail signal (Section 3.2) the skewness estimate changes sign, and the inferential conclusion with it. Under the execution timing (Section 4.4) point estimates move materially and non-monotonically in delay, while the pre-specified paired contrasts include zero. Under the symmetry of the entry rule (Section 4.5) realized performance varies across the four pre-specified rules by an order of magnitude more than the reported result, and changes sign. We do not claim a new empirical finding about factor exposure; the momentum loading reported in Section 4.7 is a property of the entry rules, not a discovery about the market.

## 2 Methodology

### 2.1 Data and Sample Construction

We sample EUR/JPY forex data from November 2015 to August 2025, sourced from Yahoo Finance (EURJPY=X). EUR/JPY is a liquid, widely traded cross rate with substantial institutional participation. The raw weekly grid begins on 6 November 2015; after rolling-statistic warmup, the analysis-ready sample runs from 8 January 2016 to 29 August 2025 and contains 504 weekly observations. Yahoo Finance quotes are indicative mid-rates and cannot be traded at directly; accordingly, all transaction cost analysis (Section 4.6) uses separate spread assumptions. The repository did not contain the raw snapshots used for the previously committed results, so this revision records retrieval times and SHA-256 hashes but does not claim bit-for-bit reproduction of that earlier run. The sample period encompasses multiple market regimes including the post-Brexit volatility spike (2016), COVID-19 pandemic shock (2020), and subsequent monetary policy divergence between the Federal Reserve, ECB, and Bank of

Japan.

**Data Frequency and Aggregation.** Alpha signals are computed daily using the specified rolling windows (5-day, 20-day, 60-day), then sampled at weekly frequency using Friday closing values. The 504 analysis-ready weekly observations are therefore end-of-week snapshots of daily-computed signals. Daily returns feed the alpha definitions and volatility estimates. Strategy returns run from the first trading-session open after one Friday signal to the first open after the next (Section 2.6); the market return used for the extreme-value diagnostic (a generalized Pareto distribution, GPD, fitted to the largest weekly return magnitudes; Section 4.8) and for the momentum factor proxy is the Friday-close-to-Friday-close change.

**Executable Sample.** Because a position is realized from the first trading-session open after its signal, the final week of the sample has no subsequent open and no realizable return. The analysis panel retains all 504 weeks, since the alpha signals are defined for every one of them, while performance statistics are computed over the 503 weeks that carry a realizable return. A week without an executable price is treated as missing, not as a zero return: recording it as zero would place an observation that does not exist into the mean, the volatility, the bootstrap and the factor regression as though it had been observed.

## 2.2 Signals

The pipeline computes five alpha signals, each capturing a distinct aspect of market dynamics:

- **Tail Alpha ( $\alpha_{\text{tail}}$ ):** Captures extreme daily price movements without discarding their direction. Formally,  $\alpha_{\text{tail},t} = \mathbf{1}_{|r_t| > q_{0.95,t}} \cdot r_t$ , where  $q_{0.95,t}$  denotes the 95th percentile of absolute daily returns over the trailing 252 trading days (with 60 observations required during warm-up). The Friday observation is retained in the weekly signal panel. This sampling rule discards most of the exceedances it detects: 105 of the 504 weeks contain at least one daily exceedance, but only 35 have one *on the Friday*, so roughly two-thirds of exceedance weeks enter the weekly panel as zeros. The resulting weekly series is far sparser than the daily construction implies, and every statistic computed on it inherits that sparsity, including its skewness, its asymmetry index and its rows in Tables 2 and 3. We retain the published construction so that the corrected results remain comparable to the published ones, and record the aliasing as a limitation without repairing it. A repair would redefine the signal and so change the specification; it would not correct it. See Section 5.3.
- **Fast Alpha ( $\alpha_{\text{fast}}$ ):** Short-term momentum signal computed as the 5-day percentage return normalized by 20-day realized volatility:  $\alpha_{\text{fast},t} = (P_t/P_{t-5} - 1)/(\sigma_{20,t}\sqrt{5})$ .
- **Pricing Alpha ( $\alpha_{\text{price}}$ ):** Mean-reversion signal measuring deviation from fair value, computed as  $\alpha_{\text{price},t} = (P_t - \text{MA}_{60,t})/\sigma_{60,t}$ , where  $\text{MA}_{60}$  and  $\sigma_{60}$  denote 60-day moving average and standard deviation.
- **Coverage Alpha ( $\alpha_{\text{cov}}$ ):** Volatility compression ratio measuring regime shifts in realized volatility:

$$\alpha_{\text{cov},t} = \frac{\sigma_{20,t}}{\sigma_{20,t-5}} - 1 \quad (1)$$

where  $\sigma_{20,t}$  denotes 20-day realized volatility computed from daily close-to-close returns. Values above zero indicate volatility expansion; values below zero indicate compression.

- **Hedge Alpha** ( $\alpha_{\text{hedge}}$ ): A correlation-based signal, defined as:

$$\alpha_{\text{hedge},t} = \rho_t^{(\text{EUR/JPY,DXY})} \times (-0.02) \quad (2)$$

where  $\rho$  is a 100-trading-day ( $\approx 20$ -week) rolling correlation of daily EUR/JPY and DXY returns (US Dollar Index, DX-Y.NYB from Yahoo Finance). The code does not use a time-varying interest-rate series: it multiplies the correlation by a fixed  $-2\%$  constant, so hedge alpha is a fixed-rate correlation proxy, not a measure of changing interest-rate regimes. Replacing the constant with a dated EUR–JPY rate differential or forward-implied measure is left for future work.

## 2.3 Asymmetry Metrics

We employ four complementary measures to characterize distributional asymmetry, each capturing distinct aspects of non-normality relevant to trading strategy design.

**Sample Skewness and Excess Kurtosis.** For a sample  $\{x_1, \dots, x_n\}$  we report the bias-corrected (adjusted Fisher-Pearson) sample skewness  $\hat{\gamma}_1$ , which measures asymmetry around the mean, and the bias-corrected sample excess kurtosis  $\hat{\gamma}_2$ , which measures tail heaviness relative to a Gaussian benchmark; Appendix B gives both estimators. Positive skewness ( $\hat{\gamma}_1 > 0$ ) indicates a right-tailed distribution with more extreme positive values, while negative skewness indicates left-tail heaviness. Positive excess kurtosis indicates a leptokurtic (fat-tailed) distribution. Under normality the standard errors are approximately  $\sqrt{6/n} \approx 0.109$  for skewness and  $\sqrt{24/n} \approx 0.218$  for excess kurtosis at our sample size ( $n = 504$ ).

**Asymmetry Index (AI)** quantifies the ratio of upside to downside semi-variance, providing a risk-management-oriented asymmetry measure:

$$AI = \frac{\text{Var}^+(X)}{\text{Var}^-(X)} = \frac{\sum_{i: x_i > \bar{x}} (x_i - \bar{x})^2 / n^+}{\sum_{i: x_i < \bar{x}} (x_i - \bar{x})^2 / n^-} \quad (3)$$

where  $n^+$  and  $n^-$  denote observations above and below the mean. Values above 1 indicate greater dispersion in positive deviations, which is relevant for assessing “lottery ticket” payoff structures (Bali et al., 2011). The index is undefined when either side is empty or the downside mean-squared deviation is zero; the implementation returns a missing value in those cases. A single observation on one side remains valid because the calculation is a mean square about the overall mean, not a subgroup sample variance.

**Positive Observation Ratio (POR)** measures the unconditional probability of positive realizations:

$$POR = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{x_i > 0} \quad (4)$$

Under normality with zero mean,  $POR = 0.5$ ; deviations indicate location-scale asymmetry or non-zero drift.

## 2.4 Statistical Tests

**Normality Tests.** We also apply three standard normality tests: Shapiro-Wilk (Shapiro and Wilk, 1965), the D’Agostino-Pearson omnibus  $K^2$  (D’Agostino et al., 1990) and Jarque-Bera (Jarque and Bera, 1980). Appendix B describes them and reports the results (Table 18).

**Skewness Significance.** We test  $H_0 : \gamma_1 = 0$  using the  $t$ -ratio  $t = \hat{\gamma}_1 / \text{SE}(\hat{\gamma}_1)$ , with the normal-theory approximation  $\text{SE}(\hat{\gamma}_1) = \sqrt{6/n} \approx 0.109$  for  $n = 504$ , which is the value used in Table 3. The exact

finite-sample formula under normality differs only in the third decimal.

**Multiple Testing Correction.** Given five alpha types and multiple test statistics, we apply the Bonferroni correction to control the family-wise error rate at the 5% level, giving a per-signal significance threshold of  $0.05/5 = 0.01$ .

## 2.5 Inference Under Serial Dependence

The skewness and return intervals reported in this paper come from two resampling schemes, applied to different objects. Skewness of the alpha signals is resampled with a *circular block* bootstrap at a fixed 13-week block: each replicate joins blocks of 13 consecutive weekly observations, drawn from random start points and wrapping around the end of the sample, until it holds 504 observations, and recomputes the bias-corrected skewness. The 95% interval is the 2.5th to 97.5th percentile of  $B = 2000$  replicates. The block is fixed because the signals are constructed from overlapping rolling windows of 5 to 252 trading days, most of them 20 to 100 days long (Table 1). Their dependence is mechanical, long, and of known length, so a fixed block of the same order as those windows is the natural choice. Strategy returns are resampled with the *stationary* bootstrap of Politis and Romano (1994) at an expected 4-week block, because they are not overlapping constructions and their dependence is short and of unknown length, which is the case the stationary bootstrap’s random block length is designed for. Only the 503 weeks with a realizable return are resampled; the final, non-executable week is excluded before resampling. The Sharpe interval is again the percentile interval of  $B = 2000$  replicates, and Table 15 reports it across expected block lengths from 2 to 13 weeks. Both schemes cover sampling variability in the weekly observations; neither covers the construction choices themselves, which Sections 3.2, 4.4 and 4.5 report as sensitivities.

## 2.6 Backtesting Framework

We evaluate the asymmetry strategy against three benchmarks, with explicit signal generation and execution rules to ensure reproducibility. Table 1 lists every free parameter.

### Asymmetry Strategy:

#### Signal Generation:

- **Entry Long:** Rolling skewness of fast alpha (20-week window) exceeds 0.75 AND current fast alpha  $> 0$
- **Entry Short:** Rolling skewness of pricing alpha (20-week window) exceeds 0.75 AND current pricing alpha  $> 0.5\sigma_{\text{pricing}}$ , where  $\sigma_{\text{pricing}}$  is the 20-week rolling standard deviation of weekly pricing alpha
- **Exit or reversal:** Preserve an open position through no-signal and same-direction-signal weeks; reverse directly when the opposing entry condition is met; otherwise exit after four realized weekly return periods. An opposing signal takes precedence if it arrives on the expiry date.
- **Simultaneous signals:** Close any open position and remain flat; do not open a new position.

#### Position Sizing:

$$\text{Position size} = \min(2.0, 1 + |AI_t - 1.0|) \quad (5)$$

where  $AI_t$  is the contemporaneous asymmetry index, evaluated at each Friday close for as long as a direction is held.  $AI_t$  is computed by Equation 3 on a trailing 20-week window of weekly fast-alpha observations, requiring a minimum of 10 observations; before that minimum is met it is undefined and

the neutral one-unit size applies. The window is the same 20-week window used for the rolling skewness entry conditions. Because  $AI_t \geq 0$ , the formula spans 1–2 gross-notional units; the previously stated 0.5-unit lower bound was unreachable. A missing AI uses the neutral one-unit size. Resizing changes the notional only: it does not open or close a holding episode, reverse a direction, or reset the holding clock.

We flag one specification choice explicitly, because it is a choice and not a correction. Repairing the exit rule (above) creates weeks in which a direction is held while no entry signal fires. The original implementation could not reach that state, because it closed any position as soon as its entry signal stopped firing, so the published specification never had to size it. Evaluating  $AI_t$  weekly, as the equation and the rebalancing convention state, and freezing the notional at entry are both extensions of the published rule to that new state. We adopt weekly evaluation as the smaller extension, since it preserves both the contemporaneous subscript in Equation 5 and the stated rebalancing frequency. Appendix F reports the frozen-notional alternative (Table 23); no conclusion in this paper depends on which is used.

*Execution:*

- Signal convention: signals use data through each Friday close, and the position is decided at that close
- Execution and return convention: the position is realized at the first trading-session open after the signal (the Monday open in an ordinary week, the next available session open after a holiday) and earns the open-to-open return to the first open after the following Friday (one lag)
- Rebalancing: weekly, re-evaluating the sizing equation at each Friday decision while a direction is held; direction changes occur only on entry, reversal, conflict, or expiry
- Exposure: positions are bounded to  $[1.0, 2.0]$  gross-notional units; values above one imply leverage relative to one unit of capital

This is the execution point the published specification named. The previously published code, and an earlier draft of this revision, priced positions at the Friday close instead; a close cannot be observed and traded at in the same instant, so the Friday close now appears only as a robustness timing. Section 4.4 applies the identical signal under four timings: Friday close, Monday open, Monday close and Tuesday open. The opening prices are indicative mid-quotes, so a fill at the open is an assumption, not an executable quote. All headline, benchmark, walk-forward, factor, cost, and data-snooping returns use the same single-lag convention.

Table 1: Complete Specification: Every Free Parameter

| Parameter                                    | Value                           | Unit                    | Governs                   |
|----------------------------------------------|---------------------------------|-------------------------|---------------------------|
| Tail exceedance quantile window              | 252                             | trading days            | Tail alpha                |
| Tail quantile minimum observations           | 60                              | trading days            | Tail alpha warm-up        |
| Tail exceedance quantile                     | 0.95                            | quantile                | Tail alpha                |
| Fast alpha return horizon                    | 5                               | trading days            | Fast alpha numerator      |
| Fast alpha volatility window                 | 20                              | trading days            | Fast alpha denominator    |
| Pricing alpha moving-average window          | 60                              | trading days            | Pricing alpha             |
| Coverage alpha volatility lag                | 5                               | trading days            | Coverage alpha            |
| Hedge alpha correlation window               | 100                             | trading days            | Hedge alpha               |
| Hedge alpha rate constant                    | -0.02                           | annualised decimal      | Hedge alpha               |
| Rolling skewness window                      | 20                              | weeks                   | Entry conditions          |
| Rolling skewness minimum observations        | 10                              | weeks                   | Entry conditions          |
| Asymmetry index window                       | 20                              | weeks                   | Equation 5 sizing         |
| Asymmetry index minimum observations         | 10                              | weeks                   | Equation 5 sizing         |
| Asymmetry index input series                 | fast alpha                      | series                  | Equation 5 sizing         |
| Entry skewness threshold                     | 0.75                            | skewness units          | Baseline strategy         |
| Threshold grid                               | 0.50, 0.75, 1.00, 1.25          | skewness units          | Sensitivity, walk-forward |
| Short-entry pricing multiple                 | 0.5                             | standard deviations     | Entry Short               |
| Maximum holding period                       | 4                               | realized weekly returns | Exit rule                 |
| Position size lower bound                    | 1                               | gross notional units    | Equation 5                |
| Position size upper bound                    | 2                               | gross notional units    | Equation 5                |
| Notional sizing                              | weekly                          | rebalance frequency     | Equation 5                |
| Execution lag                                | 1                               | weeks                   | Timing convention         |
| Execution price convention                   | first session open after signal | price                   | Timing convention         |
| Weeks with a realisable return               | 503                             | weeks                   | Sample                    |
| Pip size (JPY pairs)                         | 0.01                            | price increment         | Cost model                |
| Round-trip cost tiers                        | 0.0, 0.3, 0.7, 1.3, 2.0         | pips                    | Cost table                |
| Skewness bootstrap block length              | 13                              | weeks                   | Circular block bootstrap  |
| Skewness bootstrap replications              | 2000                            | draws                   | Circular block bootstrap  |
| Return bootstrap expected block              | 4                               | weeks                   | Stationary bootstrap      |
| Return bootstrap replications                | 2000                            | draws                   | Stationary bootstrap      |
| Data-snooping bootstrap replications         | 1000                            | draws                   | Reality Check, SPA        |
| Formal candidate universe                    | 12                              | strategies              | Reality Check, SPA        |
| Newey-West lags (full sample)                | 4                               | weeks                   | Factor regression         |
| Wild cluster bootstrap replications          | 9999                            | draws                   | Factor regression         |
| EVT exceedance threshold                     | 0.95                            | quantile                | GPD fit                   |
| EVT declustering separation                  | 5                               | weeks                   | GPD fit                   |
| Minimum episodes for reported performance    | 2                               | holding episodes        | Reporting rule            |
| Bonferroni family size (factor coefficients) | 6                               | tests                   | Factor regression         |
| Random seed                                  | 42                              | integer                 | All resampling            |
| Analysis-ready weekly observations           | 504                             | weeks                   | Sample                    |
| Analysis sample start                        | 2016-01-08                      | date                    | Sample                    |
| Analysis sample end                          | 2025-08-29                      | date                    | Sample                    |

Note: This table is generated from `analysis/specification.py`, which is the single source of these values for the analysis code. A test in `tests/test_specification.py` fails if the table and the code disagree, so a parameter cannot be changed in one without the other. Reviewer comment on the previously unstated asymmetry-index window prompted this table; the window is the “Asymmetry index window” row.

**Momentum Benchmark:** Weekly time-series momentum (Moskowitz et al., 2012): long one unit when the trailing 20-week return is positive, short one unit when it is negative, so the position is the sign of the trailing return and the strategy is always in the market once the lookback window fills.

**Mean Reversion Benchmark:** Contrarian positions on the  $z$ -score of the weekly close against its 20-week moving average: short one unit when  $z_t > 2$ , long one unit when  $z_t < -2$ , flat otherwise (positions close when  $|z|$  falls back inside the two-standard-deviation band).

**Buy & Hold:** A constant long position of one unit, entered at the start of the sample and held throughout. Its cumulative return in Table 6 compounds the weekly open-to-open EUR/JPY returns

under the same one-lag convention over the 503 weeks with a realizable return (34.63%); the close-to-close spot move from the first to the last Friday of the 504-week window, used in Table 19, is 33.53%.

The momentum and mean-reversion benchmarks are the same rules that enter the data-snooping universe, evaluated with positions applied to the following week’s return. Performance metrics include cumulative return, volatility, Sharpe and Sortino ratios, drawdown, directional holding episodes, execution legs, reversals, resizing, and turnover.

### 3 Results

#### 3.1 Asymmetry Detection

Table 2 presents distributional statistics for each alpha type, and Figure 1 plots the five distributions with their block-bootstrap skewness intervals (Appendix A reproduces both figures at larger scale).

Table 2: Alpha Asymmetry Metrics

| Alpha Type | Skew  | Kurt  | AI   | POR    |
|------------|-------|-------|------|--------|
| Tail       | −1.48 | 19.20 | 0.03 | 2.78%  |
| Fast       | 0.01  | 0.01  | 0.98 | 51.98% |
| Pricing    | −0.17 | −0.72 | 0.77 | 57.54% |
| Coverage   | 1.75  | 7.65  | 2.22 | 53.17% |
| Hedge      | 0.15  | −0.51 | 1.01 | 53.57% |

Note: Skew = bias-corrected sample skewness, Kurt = bias-corrected excess kurtosis, AI = ratio of mean squared positive and negative deviations from the overall mean, POR = positive observation ratio. All statistics use the signed weekly series ( $n = 504$ ). The earlier AI code used subgroup sample variances, which recentered each side and used  $n - 1$  denominators; the values above implement the displayed equation. An earlier version of this table reported tail-signal skewness of 5.05; that figure described the *unsigned* exceedance magnitude, which is non-negative and hence right-skewed by construction. The signed series the strategy actually trades on is reported here (currently −1.48).

*Coverage* alpha, the volatility-expansion signal, is strongly right-skewed (1.75), with high excess kurtosis (7.65) and an asymmetry index of 2.22: volatility expands abruptly and contracts gradually. Tail alpha, by contrast, has a negative point estimate (−1.48), although its dependence-robust interval includes zero (Section 3.2). Its POR of 2.78% reflects that the series is zero except at exceedances.

Signed weekly alpha signals, EUR/JPY (2016-01-08 to 2025-08-29, n=504)

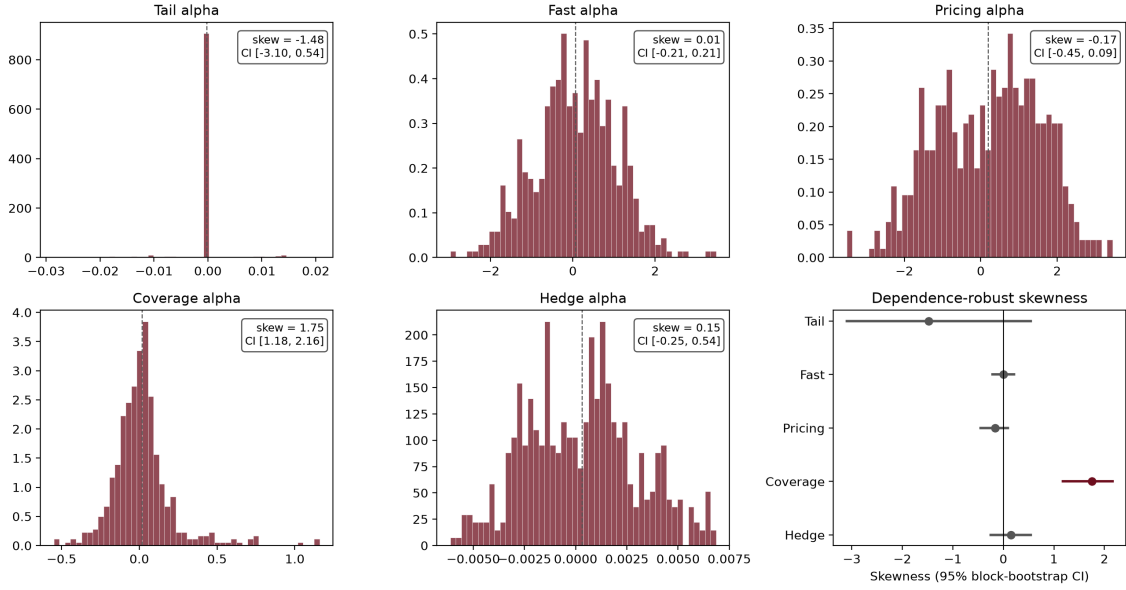


Figure 1: Alpha Asymmetry Analysis. Distributions of the five signed weekly alpha signals (EUR/JPY analysis-ready sample, January 2016–August 2025,  $n = 504$ ) with skewness point estimates and 95% circular block bootstrap confidence intervals; the final panel summarizes the dependence-robust skewness inference. Only coverage alpha’s interval excludes zero.

### 3.2 Statistical Significance

Table 3 reports the skewness tests for each alpha type.

| Alpha    | $\hat{\gamma}_1$ | $t$ -stat | Block CI      | LB $Q(4)$ |
|----------|------------------|-----------|---------------|-----------|
| Tail     | -1.48            | -13.5***  | [-3.10, 0.54] | 12.6*     |
| Fast     | 0.01             | 0.1       | [-0.21, 0.21] | 6.5       |
| Pricing  | -0.17            | -1.5      | [-0.45, 0.09] | 679.1***  |
| Coverage | 1.75             | 16.1***   | [1.18, 2.16]  | 72.6***   |
| Hedge    | 0.15             | 1.4       | [-0.25, 0.54] | 1716.2*** |

Note:  $t$ -stat tests  $H_0 : \gamma_1 = 0$  with  $SE = \sqrt{6/n} = 0.109$  ( $n = 504$ ) and assumes i.i.d. observations; Block CI = 95% circular block bootstrap confidence interval for  $\hat{\gamma}_1$  (block length 13 weeks,  $B = 2000$ ), which relaxes both the normality and the independence assumptions of the  $t$ -test; Table 4 adds an i.i.d. bootstrap interval, which relaxes normality only, so that the two effects can be separated; LB  $Q(4)$  = Ljung-Box statistic at four lags, reported per series. \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$  (unadjusted); the qualitative conclusions below apply Bonferroni correction at family size  $m = 5$ . Because pricing, coverage, and hedge alphas are built from overlapping rolling windows, their Ljung-Box statistics are large by construction and the block-bootstrap intervals are the appropriate basis for skewness inference. The Shapiro-Wilk, Jarque-Bera and D’Agostino-Pearson statistics are in Table 18 (Appendix B).

**Normality.** Four of five alpha types reject normality; for fast alpha the tests do not, which is compatibility with the null in this sample, not proof of exact Gaussianity (Table 18, Appendix B). All three normality tests assume independent observations, which the overlapping rolling constructions violate, so these rejections are indicative; dependence-robust versions exist (Bai and Ng, 2005) but are not computed here.

**Skewness.** Under the i.i.d.  $t$ -test, tail and coverage alphas show large-magnitude skewness of opposite signs. That test is not merely dependence-blind. For the sparser signals it is invalid on its own terms, because the normal-theory standard error  $\sqrt{6/n} = 0.109$  presumes a Gaussian null that tail alpha violates severely (excess kurtosis 19.20, and only 35 of 504 weekly observations are non-zero). Table 4 separates the two failures by reporting an i.i.d. bootstrap, which relaxes normality alone, beside the block bootstrap, which relaxes normality and dependence together.

Table 4: Skewness Intervals: Normal Theory, i.i.d. Bootstrap, and Block Bootstrap

| Signal   | $\hat{\gamma}_1$ | Normal-theory SE | i.i.d. bootstrap SE | i.i.d. 95% CI    | Block 95% CI    |
|----------|------------------|------------------|---------------------|------------------|-----------------|
| Tail     | -1.48            | 0.109            | 0.941               | $[-3.14, 0.62]$  | $[-3.10, 0.54]$ |
| Fast     | 0.01             | 0.109            | 0.114               | $[-0.21, 0.23]$  | $[-0.21, 0.21]$ |
| Pricing  | -0.17            | 0.109            | 0.078               | $[-0.32, -0.02]$ | $[-0.45, 0.09]$ |
| Coverage | 1.75             | 0.109            | 0.342               | $[0.98, 2.30]$   | $[1.18, 2.16]$  |
| Hedge    | 0.15             | 0.109            | 0.073               | $[0.00, 0.29]$   | $[-0.25, 0.54]$ |

Note: The normal-theory SE is the  $\sqrt{6/n}$  figure assumed by the  $t$ -test. The i.i.d. bootstrap resamples observations independently ( $B = 2000$ ), relaxing normality but not dependence. The block bootstrap (13-week circular blocks,  $B = 2000$ ) relaxes both.

The comparison is informative in two directions. For *tail alpha* the i.i.d. and block intervals are almost identical ( $[-3.14, 0.62]$  against  $[-3.10, 0.54]$ ), and the i.i.d. bootstrap standard error of 0.941 is nearly nine times the normal-theory 0.109. The width of the interval is therefore a consequence of sparsity and extreme non-normality, *not* of serial dependence; the  $t$ -test fails on this signal before dependence is considered at all. For *pricing* and *hedge* alpha the picture reverses: the block interval is substantially wider than the i.i.d. one, and in both cases it moves an interval that excludes zero to one that includes it. There the dependence correction is doing the work, as the Ljung-Box statistics of 679 and 1716 would suggest.

On the block-bootstrap yardstick, and under the primary tail construction, only *coverage* alpha survives: its interval  $[1.18, 2.16]$  excludes zero, while tail, fast, pricing and hedge alpha intervals all include zero. The tail row of that statement is construction-dependent, as the next subsection shows; the coverage row is not.

**Tail Inference Is Aggregation-Sensitive.** The tail signal is constructed by flagging exceedances in *daily* returns and then reading the flagged series at the *weekly* frequency. As Section 2.2 notes, the published rule reads the Friday observation, so 70 of the 105 weeks containing a daily exceedance enter the weekly panel as zeros. That raises a question the published specification does not answer: what is a week’s tail exposure, when the exceedances occur on days other than the one sampled?

Table 5 reports the same statistics under the published rule and under two aggregations that use every trading day.

Table 5: Weekly Tail Signal Under Three Aggregations of the Same Daily Rule

| Weekly aggregation            | Non-zero obs. | $\hat{\gamma}_1$ | Ex. kurt. | AI   | Block 95% CI   |
|-------------------------------|---------------|------------------|-----------|------|----------------|
| Friday observation (primary)  | 35            | −1.48            | 19.20     | 0.03 | [−3.10, 0.54]  |
| All days, signed sum          | 105           | +0.22            | 8.84      | 0.13 | [−0.72, 1.09]  |
| All days, largest  exceedance | 105           | −1.14            | 8.29      | 0.08 | [−1.97, −0.09] |

Note: All three apply the identical daily rule (a signed return exceeding the trailing 252-day 95th percentile of absolute returns) and differ only in how a week’s daily observations are reduced to one weekly value.  $n = 504$  weeks throughout. Block bootstrap: 13-week circular blocks,  $B = 2000$ .

The three aggregations do not agree. The point estimate changes sign between the signed-sum and the other two, and the dependence-robust interval excludes zero under one of the three. We retain the published Friday-sampled construction as primary and do not select among them. Choosing an aggregation rule after observing which one yields significance would be specification selection on outcomes, which is the practice this paper criticizes elsewhere; it is not made acceptable by the selection being ours.

Nor is either alternative self-evidently correct. The signed sum allows two exceedances of opposite sign within a week to cancel, so a violent week can register as quiet. The largest absolute exceedance allows a single day to define the week and discards every other exceedance in it. Each encodes a different and unargued claim about what weekly tail exposure means, and choosing between them requires an economic argument this paper does not make.

The finding is therefore the sensitivity itself: the tail signal’s distributional character is not robust to the aggregation rule, and is not established until weekly tail exposure is defined. That is a statement about the construction rather than about the market, and it applies equally to the published result and to both alternatives.

**The Coverage Result Does Not Depend on Aggregation.** Coverage alpha is constructed from rolling volatility and involves no sampled exceedances, so the one robust skewness result above is unaffected by the aggregation question. A robust volatility-expansion asymmetry is consistent with the broader stylized facts on asset-price distributions (Cont, 2001; Mandelbrot, 1963) and falls well short of the pervasive “alpha asymmetry” a trading strategy would want.

### 3.3 Backtest Performance

Table 6 compares the asymmetry strategy against the three benchmarks of Section 2.6 on the EUR/JPY analysis-ready sample (January 2016–August 2025). We report standard performance measures alongside risk-adjusted metrics to enable fair comparison across strategies with different return and volatility profiles.

Table 6: Strategy Performance Comparison (EUR/JPY analysis-ready sample, January 2016–August 2025)

| Strategy                  | Return  | Vol   | Sharpe | Sortino | MDD     | Episodes / Legs |
|---------------------------|---------|-------|--------|---------|---------|-----------------|
| Asymmetry                 | −0.73%  | 4.42% | 0.005  | 0.003   | −10.64% | 15 / 61         |
| Momentum (20w)            | −23.48% | 8.52% | −0.282 | −0.394  | −34.77% | 54 / 107        |
| Mean Rev. (2.0 $\sigma$ ) | −1.27%  | 3.49% | −0.021 | −0.010  | −7.63%  | 27 / 54         |
| Buy & Hold                | 34.63%  | 8.78% | 0.394  | 0.610   | −15.40% | 1 / 1           |

Note: Return = cumulative gross return; Vol = annualized volatility; MDD = maximum drawdown; all measures are computed over the 503 weeks with a realizable return. A holding episode is one continuous directional position; a direct reversal closes one episode and opens another. Execution legs count entries and exits separately, so a reversal has two legs. Turnover, resizing, and reversals are also recorded in the dated ledgers. All strategies use one timing lag. Generated by `analysis/full_pipeline.py`; an earlier version of this table reported benchmark rows (momentum −15.66%, mean reversion 34.03%, buy-and-hold 2.31%) that traced to no committed code, and the buy-and-hold figure was inconsistent with the +33.5% close-to-close EUR/JPY spot move over the sample (Table 19); all benchmark rows have been recomputed from the replication pipeline. An earlier draft of this revision counted the final, non-executable week as a zero return in Vol, Sharpe and Sortino, which changed five cells by one or two units in the last printed digit (momentum 8.51% and −0.393, mean reversion 3.48%, buy-and-hold 8.77% and 0.608).

**Risk-Adjusted Performance.** The corrected asymmetry strategy loses 0.73% gross, with a Sharpe ratio of +0.005 that is indistinguishable from zero, versus a 34.63% buy-and-hold gain. The Sharpe ratio uses the arithmetic mean weekly return, which is marginally positive (+0.04 bps, Table 10); the cumulative return compounds the same weekly returns and is negative. It holds a position in 55 of the 503 weeks with a realizable return. Its low 4.42% annualized volatility still largely reflects inactivity rather than skill. The 20-week momentum and 2 $\sigma$  mean-reversion benchmarks also lose money (−23.48% and −1.27%); only buy-and-hold is positive.

**Drawdown Analysis.** The asymmetry strategy’s maximum drawdown is −10.64%, against buy-and-hold’s −15.40%. Given that it is exposed in only 55 of the 503 weeks, losing two-thirds of what a permanently invested position lost is not evidence of risk control. Mean reversion remains more contained at −7.63%.

**Trading Frequency.** The corrected ledger records 15 asymmetry holding episodes, 61 execution legs, one direct reversal, 31 weekly resizings, and total absolute turnover of 52.00 units. The resizings more than double the leg count while adding only 5.8% to turnover, because they are small: they average 0.19 units of notional against 1.64 units at entry, the asymmetry index being a twenty-week rolling statistic that moves slowly. Momentum has 54 directional episodes because every sign reversal closes one episode and opens another; dividing position-change events by two understated this count. These definitions separate economic holding episodes from executions and turnover.

Figure 2 shows the equity curves and drawdowns.

**Statistical Significance of Returns.** The compounded annualized return is slightly negative, at −0.08%. Stationary-bootstrap inference (expected block length 4 weeks,  $B = 2000$ ; Section 2.5) gives a 95% Sharpe interval of  $[-0.73, 0.64]$ , far too wide to establish a nonzero population mean (the correction to this interval is recorded under Table 15). Failure to reject zero is not proof that the true effect is exactly absent.

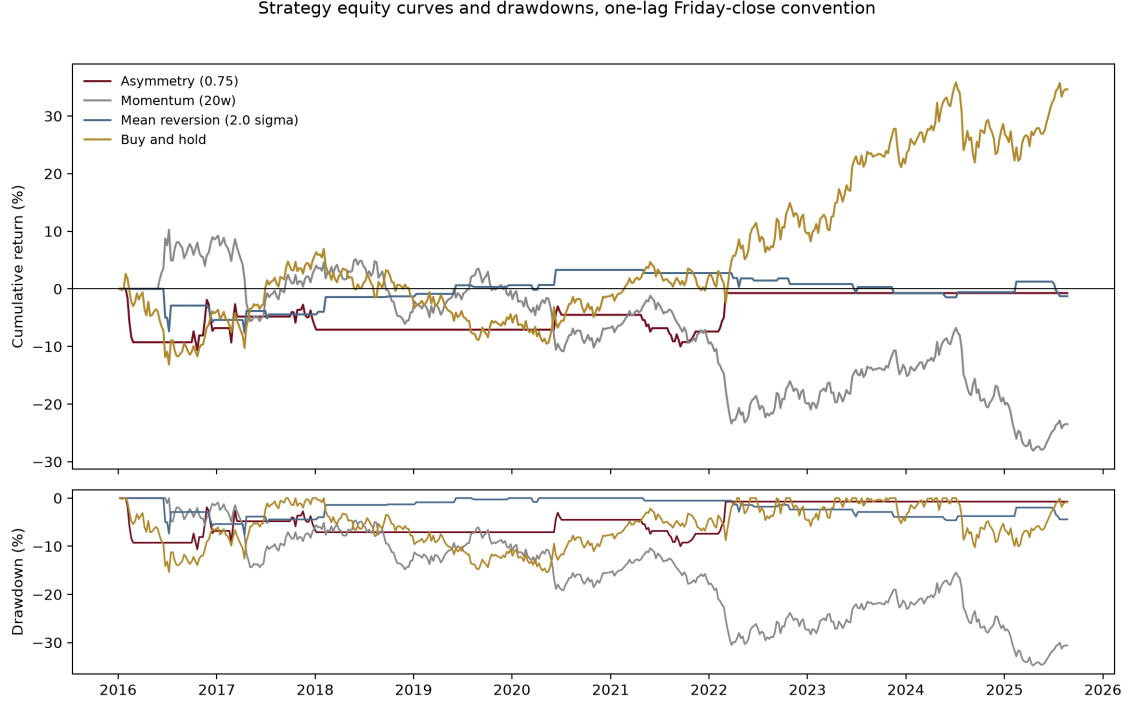


Figure 2: Strategy equity curves and drawdowns. Cumulative gross returns (top) and drawdowns (bottom) under the common one-lag convention. Generated by `analysis/full_pipeline.py`.

### 3.4 Cross-Market Check

Applied unchanged to GBP/USD, SPY and GLD, the construction and the rule do not transfer cleanly: EUR/JPY’s negative tail-skew point estimate is positive in all three other markets while coverage alpha’s right skew recurs, and the strategy loses 14.45% on GBP/USD, gains 14.71% on SPY while trailing buy-and-hold by roughly 280 percentage points, and loses 13.03% on GLD (Appendix C, Table 19). These exploratory transfers lack pair-specific costs and dependence-aware inference, so they do not establish effects in any market. An earlier version of this check reported aggregated “MR/TF/HAT” skewness categories and returns that traced to no committed code; they have been replaced.

## 4 Robustness Analysis

This section stress-tests the in-sample results of Sections 3.2 and 3.3, addressing four questions: (1) Do results hold out-of-sample? (2) Are patterns stable across market regimes? (3) What do trading costs do to an already negative return? (4) Does the strategy’s return stream proxy for known FX risk factors?

### 4.1 Out-of-Sample Validation

A primary concern with any trading strategy is in-sample overfitting. Our main results use the full January 2016–August 2025 analysis-ready sample, which may inadvertently optimize signal construction to historical patterns that do not persist. We address this through walk-forward analysis, in which the strategy’s free parameter is chosen on past data only; on the broader problem of evaluating technical analysis statistically, see [Lo et al. \(2000\)](#).

We implement an expanding-window walk-forward: the entry threshold is selected from the grid  $\{0.50, 0.75, 1.00, 1.25\}$  by Sharpe ratio on all analysis-ready data through year  $t - 1$  (beginning with 2016–2017), then the strategy runs untouched on calendar year  $t$ , rolling forward annually through 2025.

All signal construction uses trailing windows only, so no test-year information enters signal values; the walk-forward isolates the threshold-selection step, which is the strategy’s only free parameter. Table 7 reports what that procedure did.

Table 7: Out-of-Sample Walk-Forward: Selected Thresholds and Activity

| Test Year               | Training Window | Selected Threshold     | New Holding Episodes |
|-------------------------|-----------------|------------------------|----------------------|
| 2018                    | 2016–2017       | 1.25                   | 0                    |
| 2019                    | 2016–2018       | 1.25                   | 0                    |
| 2020                    | 2016–2019       | 1.25                   | 1                    |
| 2021                    | 2016–2020       | 1.25                   | 0                    |
| 2022                    | 2016–2021       | 1.25                   | 0                    |
| 2023                    | 2016–2022       | 1.25                   | 0                    |
| 2024                    | 2016–2023       | 1.25                   | 0                    |
| 2025                    | 2016–2024       | 1.25                   | 0                    |
| <b>Eight test years</b> | <b>—</b>        | <b>1.25 throughout</b> | <b>1</b>             |

Note: The threshold is selected by Sharpe ratio on data through the prior year, then applied on one sequential out-of-sample state path. An episode is a newly opened directional holding, not a position-change pair. 2025 ends in August. No out-of-sample return, Sharpe ratio or hit rate is reported in this table, by design. With one episode there is no sample over which to compute them; see the text. Generated by `analysis/full_pipeline.py`; an earlier version of this table reported 141 pooled trades and could not be reproduced from the strategy specification. The corrected count of newly opened out-of-sample episodes is one.

**Why No Out-of-Sample Performance Figures Appear.** The corrected training windows select the 1.25 threshold in every year. Across eight out-of-sample years the strategy opens *one* holding episode, in 2020, and is flat in all other test years. One episode is not a small sample; it is a single realization, and the quantities an out-of-sample table would normally carry are not estimable from it. A Sharpe ratio, a hit rate and an annualized return are sample statistics that presuppose a distribution of outcomes to average over; computed from one holding episode they describe the path that happened and carry no information about the path that would be expected. A hit rate of 50% on such a sample means one profitable week and one unprofitable week, and an annualized figure extrapolates a single episode across a decade. Reporting them, even hedged, invites the comparison they cannot support.

We therefore report the selected threshold and the episode count, which describe what the selection procedure did, and decline to report out-of-sample performance. The finding of this section is the activity count itself: disciplined by its own training data, the strategy is inert. Unlike a return computed from a single episode, that result does not depend on how the episode happened to turn out. The replication pipeline enforces this rule in code: it flags the walk-forward as not supporting inference whenever fewer than two out-of-sample episodes occur, and records the suppressed quantities in `analysis/full_pipeline_results.json` so the decision can be audited.

## 4.2 Subsample Stability

The sample includes the COVID-19 shock, and we test whether asymmetry patterns persist across distinct market environments using subsample analysis.

Table 8 partitions the sample into subperiods.

Table 8: Subsample Asymmetry Stability

| Subsample                    | $N$ | Tail Skew | Fast Skew | Price Skew | Strategy Ret. |
|------------------------------|-----|-----------|-----------|------------|---------------|
| Pre-COVID (2016–2019)        | 208 | −3.40     | −0.05     | 0.12       | −7.08%        |
| COVID Shock (2020)           | 52  | 5.02      | 0.09      | −0.07      | 2.77%         |
| Post-COVID (2021–2025)       | 244 | −0.73     | 0.05      | −0.38      | 3.96%         |
| Low VIX ( $VIX < 20$ )       | 342 | −3.25     | 0.04      | −0.20      | −1.69%        |
| High VIX ( $VIX \geq 20$ )   | 162 | 0.62      | −0.06     | −0.10      | 0.98%         |
| Rate Hike Regime (2022–2025) | 191 | −0.71     | −0.13     | −0.53      | 7.21%         |

Note: The strategy is run once on the complete chronological calendar; realized weekly returns are then attributed to rows in each regime. VIX is the mean of daily closes within each Friday-ending week. No stateful strategy is rerun on filtered, nonconsecutive dates. Skewness uses the observations in each displayed subsample. The 2020 return of 2.77% is identical under both position-sizing specifications of Section 2.6, and is the same figure as the lone walk-forward episode of Table 7: 2020 contains a single holding episode whose notional the sizing rule never revised, so weekly and frozen sizing coincide on it. It is not a figure left un-updated.

**Temporal Stability.** Tail skewness is strongly negative pre-COVID (−3.40), positive in 2020 (+5.02), and moderately negative thereafter (−0.73). These point estimates are based on small, sparse exceedance samples and are descriptive. Fast skew remains near zero; pricing skew is more negative after 2020.

**Regime Conditioning.** The returns attributed to low-VIX and high-VIX weeks are −1.69% and +0.98%, and they compound to the full-sample −0.73%. These are ex post attributions of a single strategy run, not separately tradable conditioned strategies, so they describe where gains and losses occurred and do not establish a usable VIX timing rule.

### 4.3 Entry Threshold Sensitivity

The baseline skewness threshold (0.75) deserves scrutiny: is it arbitrary or robustly selected? Table 9 reports strategy performance across alternative thresholds, holding other parameters constant.

Table 9: Sensitivity to Entry Threshold

| Threshold | Episodes | Exposed Weeks | Return                              | Sharpe | Hit Rate |
|-----------|----------|---------------|-------------------------------------|--------|----------|
| 0.50      | 25       | 89            | −10.55%                             | −0.19  | 49.4%    |
| 0.75      | 15       | 55            | −0.73%                              | 0.00   | 49.1%    |
| 1.00      | 4        | 16            | −1.25%                              | −0.06  | 50.0%    |
| 1.25      | 1        | 4             | <i>single episode: not reported</i> |        |          |

Note: Returns use the full asymmetry strategy specification of Section 2.6 with weekly position sizing. Activity is placed first because it is the stable feature of the grid: raising the threshold monotonically reduces the number of episodes and the weeks of exposure, while the return does not vary monotonically. The 1.25 row reports no performance figures, for the same reason as Table 7 and in fact from the same episode: one holding episode supplies no sample over which a Sharpe ratio or a hit rate can be computed. The replication pipeline applies this rule to every such statistic in every table, and records the suppressed values in `analysis/full_pipeline_results.json`.

**Threshold Selection.** What the grid shows unambiguously is that activity falls monotonically as the threshold rises: 25 episodes and 89 exposed weeks at 0.50, down to a single episode and four exposed weeks at 1.25. Return and Sharpe do not move monotonically: the return improves from −10.55% to −0.73% and falls back to −1.25%. Every threshold with enough episodes to evaluate loses money. The 1.25 threshold is the same lone 2020 episode that the walk-forward selects, and it is reported here only as an activity count, for the reason given in Section 4.1: a single episode cannot distinguish an edge

from a coin flip. Selecting it after seeing the full sample would in any case be data snooping rather than robustness. The published 0.75 baseline remains the reported result.

#### 4.4 Execution-Timing Sensitivity

The execution convention is a choice the original specification stated but did not argue for, and the previously published code departed from it. Table 10 reports four timings, each applied to the *identical* Friday-close signal and differing only in when the resulting position is established and therefore which one-week return it earns.

Table 10: Execution-Timing Grid: Identical Signals, Four Entry Points

| Entry timing                    | Cum. return | Mean weekly | Sharpe | MDD     | Net at 2.0 pips |
|---------------------------------|-------------|-------------|--------|---------|-----------------|
| Friday close                    | −6.64%      | −1.21 bps   | −0.154 | −12.56% | −7.02%          |
| Monday open (reported baseline) | −0.73%      | +0.04 bps   | +0.005 | −10.64% | −1.14%          |
| Monday close                    | −0.92%      | 0.00 bps    | 0.000  | −10.72% | −1.32%          |
| Tuesday open                    | −8.03%      | −1.50 bps   | −0.185 | −17.19% | −8.40%          |

Note: Common sample of 502 weeks, being every week for which all four entry points and their one-week-forward counterparts exist; two weeks are dropped, the same two for every timing, so no differential attrition can be mistaken for an execution effect. Signals, entry and exit rules, the holding clock and position sizing are identical throughout. Costs at the tiers of Table 12. Generated by `analysis/full_pipeline.py`, as are the paired contrasts and the weekend-gap test reported in the text. An earlier draft of this revision took those from a script that was never committed; the pipeline reproduces its three intervals, and its point estimates of 0.64 and −0.76 points become 0.63 and −0.77.

The four span 7.3 percentage points of cumulative gross return, and the ordering is not monotone in execution delay: the latest timing tested is worse than the earliest, while two timings hours apart differ by less than a fifth of a point.

A paired moving-block bootstrap finds no pairwise difference distinguishable from zero. The contrast is the difference in annualized return, later timing minus earlier. Each of  $B = 2000$  replicates draws fixed four-week blocks of weeks and applies the same resampled weeks to every timing, so the comparison is paired, and the 95% interval is the 2.5th to 97.5th percentile of the replicates. The difference between the published Monday-open convention and the Friday close is 0.63 percentage points, 95% CI [−0.67, 2.03]; between Monday open and Monday close, −0.02 points, CI [−0.08, 0.04]; between Monday close and Tuesday open, −0.77 points, CI [−2.36, 0.45]. These conclusions are unchanged at eight- and thirteen-week blocks and under a stationary bootstrap (Table 20 in Appendix D).

**What was specified in advance, and what was not.** The Friday-close and Monday-open results were already in hand when this exercise began, and we do not present them as prospectively specified. *Before* computing the Monday-close and Tuesday-open variants, we specified the three primary contrasts above, the interpretation rule attached to them, the bootstrap procedure and block lengths, and the common-sample requirement, and committed that plan to the replication repository. All other pairwise comparisons among the four timings are secondary and are handled under the pre-declared family-wise procedure; none of the three secondary simultaneous intervals excludes zero (Table 20).

The interpretation rule was that the difference would be treated as weekend-specific only if the contrasts crossing no weekend failed to move comparably. One of them, Monday close to Tuesday open, moves as much as the weekend-crossing contrast does. We therefore abandon the weekend-gap explanation, as the rule required. We offer no replacement mechanism: proposing a second explanation after inspecting the same data is the specification search this paper criticizes elsewhere. We also tested directly whether the entry rule predicts the weekend gap a position is about to bear, regressing the move

from the decision Friday’s close to the next session open on the signed position over the 55 decision weeks with a position, with CR2 errors clustered by the 15 holding episodes. We found no detectable effect in this sample (slope  $-1.4$  basis points per unit of position, wild cluster bootstrap  $p = 0.58$ ), but the test could resolve only an effect several times larger than the point estimate: in a simulation that resamples episodes and injects a known slope, the test reaches 80% power at about 8.0 basis points per unit, 5.8 times the point estimate.

What the grid establishes is narrow: the headline result is sensitive to a design choice the original specification did not argue for, at a magnitude this sample cannot resolve.

Execution timing is one of three unargued choices whose statistical behavior differs, alongside the tail aggregation and the entry-rule symmetry (Section 1 and Table 17); Section 4.5 reports the last.

#### 4.5 Entry-Rule Symmetry

The long and short legs of the entry rule differ in four respects: the skewness series that gates them, the confirmation series, the confirmation threshold, and the direction of response. A positive fast alpha opens a long position; a positive pricing alpha opens a short one. The rule therefore follows strength in one signal and fades it in the other, and the original specification argues for none of this.

Collapsing the asymmetry is underdetermined, because it requires choosing which of the two economics to keep. Three symmetrizations were specified in advance, together with the metrics, the cost tier, an exposure guard, and a reporting rule fixing that all three would be disclosed whatever they returned; that specification was committed before any of them was computed.<sup>1</sup>

Table 11: Entry-Rule Symmetry: The Published Hybrid and Three Pre-Specified Symmetrizations

| Entry rule                              | Gross  | Net 2.0p | Sharpe | MDD     | Eps. | Weeks | bps/week |
|-----------------------------------------|--------|----------|--------|---------|------|-------|----------|
| Published hybrid (reported baseline)    | -0.73% | -1.14%   | 0.005  | -10.64% | 15   | 55    | 0.4      |
| Pure-fast (both legs, fast alpha)       | -3.48% | -3.97%   | -0.067 | -12.05% | 16   | 44    | -6.1     |
| Pure-pricing (both legs, pricing alpha) | 5.57%  | 5.21%    | 0.178  | -11.97% | 14   | 51    | 11.8     |
| Equal-threshold hybrid                  | -1.65% | -1.96%   | -0.029 | -10.83% | 12   | 43    | -2.4     |

Note: All four rules use the identical sample of 504 weeks, the identical sizing, exit and execution conventions, and differ only in the entry condition. “Weeks” counts in-position weeks and “bps/week” is the mean gross weekly return over those weeks, reported for every rule under the pre-specified exposure guard so that a rule which is out of the market more often cannot be read as safer. Net returns are quoted at 2.0 pips round-trip, the widest tier in the paper’s cost table. Generated from `analysis/entry_symmetry_results.json`, which also holds the Friday-close runs of the four rules quoted in the text.

Realized performance varies materially across the four pre-specified entry rules (Table 11), and the variation includes a change of sign. The published hybrid loses 0.73% gross; pure-fast loses 3.48%; the equal-threshold hybrid loses 1.65%; and pure-pricing gains 5.57% gross, 5.21% net of the widest cost tier. The spread is 9.04 percentage points, against a published-hybrid loss of 0.73%, an order of magnitude larger than the result it surrounds.

**What the positive variant does and does not establish.** Pure-pricing is a pre-specified variant, not a rule found by searching. Its return is reported because the pre-specification, committed to the repository before the execution convention was decided and before any of these figures existed, required

<sup>1</sup>`docs/PREREGISTRATION_ENTRY_SYMMETRY.md`. Two structural predictions recorded in that document were wrong and are left standing there: pure-fast was predicted to hold positions in more weeks than the published rule and holds fewer, and pure-pricing was predicted to be less exposed than pure-fast and is more so. Both errors share a cause, namely that the published rule draws entries from the union of two skewness gates while each collapsed variant has only one. Section 6 of that document also asserts that the variants cannot be distinguished statistically. No paired inferential comparison among the four rules was pre-specified or run, so that statement is untested, neither established nor rejected; no conclusion reported here relies on it, and it is not repeated.

all three symmetrizations to be disclosed whatever they produced. It is *not* offered as a preferred strategy, and no comparative inference among the four was pre-specified or run, so nothing here establishes that pure-pricing has positive expected alpha. What it establishes is narrower: the sample does not support a stable performance conclusion across the pre-specified entry-rule constructions.

The claim this paper made in an earlier draft—that the economic failure is robust to how the entry asymmetry is resolved—is no longer supportable, and we withdraw it rather than qualify it. Four defensible readings of one rule, differing only in a design choice the original specification never argued for, produce outcomes spanning from a 3.48% loss to a 5.57% gain.

Exposure does not explain the spread. Pure-fast and pure-pricing hold positions in fewer weeks than the published rule (44 and 51 against 55), and the ordering is the same per exposed week as it is cumulatively:  $-6.1$  and  $+11.8$  basis points against the published rule’s  $+0.4$ . A rule that is out of the market more often is not thereby made to look better here.

Equalizing the confirmation threshold worsens realized performance under both execution conventions, by 1.22 percentage points under Friday close and 0.92 under Monday open. This is one comparison and is not evidence that the published threshold was tuned; its direction is merely consistent with what a specification-search concern would predict.

These are sensitivity exhibits. No variant is offered as a replacement rule, and realized performance was pre-specified not to determine which symmetrization is treated as defensible.

#### 4.6 Transaction Cost Sensitivity

Backtests without transaction costs overstate implementable returns. We analyze cost sensitivity across scenarios ranging from zero-cost (theoretical benchmark) to retail-level spreads.

The cost tiers in Table 12, from 0.3 to 2.0 pips round trip, are stylized assumptions, not measured spreads: the sample has no executable EUR/JPY quotes. Bid-ask spreads in the interdealer FX market rise with volatility and trading volume and widen at times of heightened risk, while the spreads dealers quote to customers follow a different pattern (King et al., 2013). Spread widening during volatility spikes and the cost of crossing the spread on each trade can therefore erode returns, particularly for higher-frequency strategies.

Table 12: Transaction Cost Analysis

| Cost Scenario        | Round-Trip Cost (pips) | Net Return | Net Sharpe |
|----------------------|------------------------|------------|------------|
| Zero Cost (Baseline) | 0.0                    | $-0.73\%$  | 0.005      |
| Prime Brokerage      | 0.3                    | $-0.79\%$  | 0.003      |
| Institutional        | 0.7                    | $-0.87\%$  | 0.001      |
| Retail (Tight)       | 1.3                    | $-0.99\%$  | $-0.001$   |
| Retail (Wide)        | 2.0                    | $-1.14\%$  | $-0.005$   |

Note: Each execution leg is charged half the round-trip pips in proportion to absolute notional change at that row’s indicative price. Financing is excluded; because exposure can exceed one unit and lasts 55 weeks, its omission is a limitation, not assumed immaterial. No break-even row appears because the zero-cost return is already negative (see the text). Spread is charged in proportion to notional traded, with no fixed per-order component; see the discussion below. Generated by `analysis/full_pipeline.py`; an earlier version of this table showed net return *rising* with costs at low tiers, an inconsistency in the carry treatment across rows. Net return is now monotonically decreasing in cost, as it must be.

**Cost Components.** We model spread and impact per dated execution leg, charged in proportion to the notional actually traded. Financing/carry is excluded uniformly because no suitable dated rate or forward series is present; the omission may matter for leveraged positions and should be addressed with

a dated rate or forward series in future work.

**There Is No Break-Even Cost.** The published version of this paper reported a break-even round-trip cost of 19.2 pips, and read that figure as reassurance: costs would have to be implausibly wide before they consumed the strategy’s edge. That number was meaningful only for as long as the gross return was positive. The corrected specification has no break-even cost at all. The strategy returns  $-0.73\%$  before any cost is applied, so there is no positive cost level at which it crosses zero; it begins below zero and every cost tier moves it further away.

The earlier framing invited the reader to ask whether frictions were large enough to matter, which is the right question only when a gross edge exists to be eroded. Under the corrected specification the question does not arise. Costs are not the reason the strategy fails, and they are not exonerated either: they are downstream of a backtest that has already failed.

**Cost Interpretation.** Execution costs make an already negative result more negative, monotonically and by little: gross  $-0.73\%$  becomes  $-1.14\%$  at two pips round-trip, a drag of 0.41 percentage points. Under the previous execution convention the gross loss dwarfed that drag; it no longer does, and the cost tier is now of the same order as the gross result itself.

**What This Cost Model Cannot See.** The modest size of the cost drag is partly a property of the model and should not be over-read. Spread is charged strictly in proportion to notional traded, with no fixed or minimum per-order component. The corrected specification generates 31 weekly resizings averaging 0.19 units of notional. These are small orders, the population on which a fixed per-order charge falls hardest. A realistic commission schedule with any per-ticket element would put a positive price on those 31 orders, and the drag reported above is correspondingly a lower bound. It would not change the sign of the result, but “costs are immaterial” is a claim this model is not equipped to make.

#### 4.7 Factor Attribution

Does the strategy’s return stream proxy for known FX risk factors? Prior work documents systematic risk premia in currency markets associated with carry (Lustig et al., 2011), momentum (Menkhoff et al., 2012), and the dollar factor (Verdelhan, 2018), with Lettau et al. (2014) showing that downside risk conditioning explains a substantial portion of currency risk premia across asset classes. If the strategy simply loads on them, any residual return would be factor timing, not an asymmetry effect.

We regress asymmetry strategy returns on proxy FX factors:

$$R_{\text{asym},t} = \alpha + \beta_1 \cdot \text{Carry}_t + \beta_2 \cdot \text{Mom}_t + \beta_3 \cdot \text{Dollar}_t + \varepsilon_t \quad (6)$$

where  $\alpha$  is the regression intercept, the residual return in the usual sense of the factor literature and distinct from the alpha signals of Section 2.2, and the factors are single-market proxies constructible from our data, not the published cross-sectional factors: Carry is the average weekly spot return of the classic carry crosses AUD/JPY and NZD/JPY (a spot-only proxy for the Lustig et al., 2011 carry factor, omitting interest accrual), Mom is the return of a 12-week time-series momentum rule on EUR/JPY itself (in the spirit of Menkhoff et al., 2012), and Dollar is the DXY weekly return (proxying Verdelhan, 2018). Constructing the published factors requires a G10 forward-rate cross-section we do not use; the proxies capture the same directional exposures at the single-pair level, and we label them as proxies throughout. Because the strategy is out of the market most weeks, we estimate the regression twice: on the full weekly sample (where the dependent variable is structurally zero in out-of-position weeks) and on in-position weeks only, whose count is the effective sample size. Table 13 reports both.

Table 13: Factor Attribution Regression (HAC-Corrected)

| Variable               | Full sample ( $n = 503$ ) |           | In-position weeks ( $n = 55$ ) |           |
|------------------------|---------------------------|-----------|--------------------------------|-----------|
|                        | Coefficient               | $t$ -stat | Coefficient                    | $t$ -stat |
| Intercept ( $\alpha$ ) | 0.00000                   | 0.01      | -0.00095                       | -0.45     |
| Carry ( $\beta_1$ )    | 0.0159                    | 0.63      | 0.218                          | 1.11      |
| Momentum ( $\beta_2$ ) | -0.0375                   | -1.71     | -0.652                         | -2.33*    |
| Dollar ( $\beta_3$ )   | 0.0607                    | 1.59      | 0.622                          | 2.64*     |
| $R^2$                  | 0.013                     |           | 0.170                          |           |
| $F$ -statistic         | 1.47                      |           | 3.47                           |           |

Note: Full-sample column uses Newey-West HAC standard errors with 4 lags. The in-position column uses CR2 standard errors clustered by holding episode (55 weeks in 15 clusters) with Bell-McCaffrey (Satterthwaite-type) degrees of freedom, because those weeks are non-contiguous (they are drawn from separate episodes spread across a decade) and a lag-based correction presumes consecutive observations. The reported  $p$ -values for momentum come from a restricted wild cluster bootstrap; see Table 14. \*  $p < 0.05$  (unadjusted). The full-sample  $F$  is the Wald statistic for the three slopes under the same Newey-West covariance, so it does not follow from  $R^2$ ; the in-position  $F$  is the classical OLS statistic, which ignores the clustering and is shown for reference only. The full sample includes the flat weeks in which the dependent variable is structurally zero; the in-position sample of 55 weeks is the effective exposure sample and the economically interpretable one. The intercept is not significant in either specification. Momentum is negative in both specifications but is no longer the most strongly loading factor. Compared like for like, on CR2 standard errors from the same estimator and clustering, the dollar proxy carries the larger  $t$ -statistic on in-position weeks (+2.64, CR2  $p = 0.034$ ) than momentum does (-2.33, CR2  $p = 0.042$ ). The reported momentum  $p$ -value of 0.0506 comes from the restricted wild cluster bootstrap and is not comparable with either CR2 figure; the bootstrap was computed for momentum alone. On the full sample momentum reaches only  $p = 0.087$ . The sample-selection problem described below applies to every coefficient in the in-position regression alike, so no coefficient in that column is offered as an identified exposure. Under the earlier specification the same coefficient was -0.007 ( $t = -0.37$ ,  $p = 0.71$ ) on the full sample and -0.247 ( $t = -0.51$ ,  $p = 0.61$ ) in position; the change is a consequence of correcting the holding rule and the execution lag, not of altering the regression. Generated by `analysis/full_pipeline.py`; an earlier version of this table reported a marginally significant intercept of 21 bps weekly (10.9% annualized) with  $F = 20.8$ , which could not be reproduced and was internally inconsistent with its own  $R^2$  and with the 3.60% cumulative backtest return. Those figures describe the state of the paper at that time; the corresponding values here are an intercept of +0.04 bps weekly (+0.00000351 in weekly decimal return) and  $F = 1.47$ , against a cumulative backtest return of -0.73%.

**No Evidence of Positive Residual Alpha.** The intercept is statistically indistinguishable from zero in both specifications: essentially zero on the full sample and negative on in-position weeks ( $t = 0.01$  full sample;  $t = -0.45$  on in-position weeks). This is evidence against a positive residual-alpha claim, not proof that the population intercept is exactly zero.

**What the In-Position Regression Can and Cannot Show.** Restricting the regression to in-position weeks is not a neutral choice of sample. Those weeks are selected by the strategy's own entry rules, and those rules are functions of the same price series from which the momentum factor is built. The short leg fires when pricing alpha is high, that is, after price has risen relative to its sixty-day average, and a twelve-week time-series momentum rule is long in precisely those states. The sample and the regressor are jointly determined by the entry condition.

The negative loading on in-position weeks ( $\beta_2 = -0.652$ , CR2 interval  $[-1.27, -0.03]$ , wild cluster bootstrap  $p = 0.0506$ ) should therefore be read as a description of what the entry rules mechanically produce, not as an estimate of a factor exposure the strategy carries independently of them. A rule whose short leg fades medium-term strength on the same price series will load negatively on momentum during the weeks it is active whether or not any factor relationship exists in the underlying returns. We report the coefficient because it characterizes the rule. We do not interpret it as explaining the strategy's losses, and we do not present it as an empirical finding about the market.

The full-sample regression is reported alongside as an economically relevant description of the complete strategy return stream: it is what an investor holding the rule would have experienced, flat weeks included. There the loading is  $\beta_2 = -0.038$  ( $p = 0.087$ ), attenuated by the 448 weeks in which the dependent variable is structurally zero, and it does not meet the paper’s Bonferroni-adjusted threshold. It does not solve the identification problem either. Including the inactive weeks changes what is being described; it does not make the active weeks independent of the entry rule that selected them. The intercept is indistinguishable from zero in both specifications, and neither supports a claim about residual alpha.

**Inference Was Tightened Three Times, and That Was Not the Binding Problem.** The estimated significance of this loading moved substantially as the inferential treatment improved:  $p = 0.00019$  under the Newey-West treatment first applied, and  $p = 0.0091$  once errors were clustered by holding episode (ordinary cluster-robust errors). Both were computed while positions were still priced at the Friday close; the rule, the sizing, the 55-week sample and the momentum proxy were otherwise the same. Under CR2 with a restricted wild cluster bootstrap appropriate to fifteen clusters, the two figures no longer agree: the CR2 interval  $[-1.27, -0.03]$  excludes zero at  $p = 0.042$  while the bootstrap returns  $p = 0.0506$ . The 5% line now falls between the two methods. Each step was a correction, and the first two were withdrawn as inapplicable rather than merely superseded. Fifteen clusters is few, and ordinary cluster-robust errors are asymptotic in the number of clusters rather than the number of observations, so they over-reject when clusters are scarce. The reported figures therefore use the bias-reduced CR2 estimator of [Bell and McCaffrey \(2002\)](#) with Bell–McCaffrey degrees of freedom (10.2 here, against a naive 14) and take the  $p$ -value from a restricted wild cluster bootstrap- $t$  with Rademacher weights ([Cameron et al., 2008](#)). Even these are approximations at fifteen clusters, and the  $p$ -value is indicative only. Table 14 reports all of it.

Table 14: Momentum Loading Under Small-Cluster Inference

| Inference                      | $\hat{\beta}_2$ | SE    | $t$   | $p$           | Role                        |
|--------------------------------|-----------------|-------|-------|---------------|-----------------------------|
| Wild cluster bootstrap- $t$    | −0.652          | —     | −2.33 | <b>0.0506</b> | <b>Primary</b>              |
| CR2, Bell–McCaffrey dof = 10.2 | −0.652          | 0.280 | −2.33 | 0.042         | Standard error and interval |
| HC3                            | −0.652          | 0.346 | −1.88 | 0.060         | Robustness                  |
| Newey-West HAC, 4 lags         | −0.652          | 0.251 | −2.60 | 0.0094        | <i>Withdrawn</i>            |

Note: 55 in-position weeks in 15 holding episodes. The point estimate is identical throughout the table; only the inference differs. Ordinary cluster-robust errors are asymptotic in the number of clusters, and fifteen is few, so CR2 supplies the standard error and the 95% interval  $[-1.27, -0.03]$ , and a restricted wild cluster bootstrap- $t$  with Rademacher weights ( $B = 9,999$ , null imposed) supplies the primary  $p$ -value. The Newey-West row is not an alternative specification. It shows the withdrawn Newey-West treatment recomputed on the corrected specification, so that the withdrawn method is identifiable; the earlier draft reported  $p = 0.00019$  for the same treatment with positions priced at the Friday close. A lag-based autocorrelation correction presumes consecutive observations, which these are not, and it is inapplicable here.

The selection problem described above is a different kind of objection. Improved inference can correct the uncertainty attached to a coefficient conditional on a specification; it cannot confer an independent interpretation on a sample selected by rules mechanically related to the regressor. No standard-error treatment, however careful, would have converted this coefficient into evidence about a factor exposure. The sequence is recorded because the distinction is easy to lose: three rounds of tightening made the estimate less impressive, and only the fourth objection made it a different kind of object.

Two further qualifications bound whatever descriptive content remains. Mom is a twelve-week time-series momentum rule on EUR/JPY itself, not the published cross-sectional currency momentum factor

of Menkhoff et al. (2012), so a strong loading of a single-pair strategy on a single-pair momentum rule is neither surprising nor transferable. And the three proxies together leave most of the variation in in-position returns unexplained ( $R^2 = 0.170$ , Table 13).

#### 4.8 Weekly-Return Tail Distribution

As a separate diagnostic, Appendix E fits a generalized Pareto distribution to the cluster maxima of absolute Friday-close-to-Friday-close EUR/JPY returns above their 95th percentile, a different sample from tail alpha. The shape estimate,  $\hat{\xi} = -0.25$  with 95% CI  $[-1.49, 0.27]$  from 20 cluster maxima, is too imprecise to establish either heavy tails or exponential decay, at every declustering separation tried; clustering is weak (extremal index  $\theta = 0.83$ , CI  $[0.60, 1.00]$ ), and the fit provides no affirmative evidence for the lottery-like right tail the trading story needs (Tables 21 and 22). An earlier version reported an extremal-index interval of  $[0.42, 0.63]$  that excluded its own point estimate; it came from a biased index-resampling scheme and has been replaced.

#### 4.9 Multiple Testing Corrections

With five alpha types examined across multiple statistical tests, data-snooping concerns arise: apparent significance may reflect chance discoveries from multiple hypothesis testing. We address this through both formal corrections and bootstrap-based reality checks.

**Bonferroni Corrections.** Table 3 applies plain Bonferroni correction at family size  $m = 5$  (one skewness test per alpha type; a per-signal threshold of 0.01 for a 5% family-wise level). At the corrected threshold, four of five alpha types reject *normality* (Table 18), but the asymmetry inventory is far thinner: under the primary construction only coverage alpha’s skewness survives both the correction and, more demanding, the block-bootstrap dependence adjustment. Coverage is the only signal for which that statement does not depend on a sampling choice. Fast alpha, for which the tests do not reject normality (Section 3.2), serves as a control.

**Return Bootstrap Block Length.** The resampling schemes are described in Section 2.5. Table 15 reports the Sharpe interval of Section 3.3 across expected block lengths from 2 to 13 weeks; every interval contains zero, so no conclusion rests on the choice.

Table 15: Sensitivity of the Sharpe Interval to Expected Block Length

| Expected block length | 95% Sharpe interval |
|-----------------------|---------------------|
| 2 weeks               | $[-0.733, 0.600]$   |
| 4 weeks (reported)    | $[-0.723, 0.664]$   |
| 8 weeks               | $[-0.654, 0.608]$   |
| 13 weeks              | $[-0.755, 0.585]$   |

Note: Stationary bootstrap,  $B = 1000$ , applied to the corrected weekly strategy returns over the 503 weeks with a realizable return. Every interval contains zero. The 4-week row uses  $B = 1000$  and therefore differs slightly from the interval reported in Section 3.3, which uses  $B = 2000$ . Generated by `analysis/full_pipeline.py`. An earlier draft of this revision reported narrower intervals in this table and in Section 3.3 ( $[-0.51, 0.49]$  there;  $[-0.484, 0.505]$ ,  $[-0.451, 0.496]$ ,  $[-0.370, 0.558]$  and  $[-0.236, 0.593]$  here). They were computed with the final, non-executable week left in the resampled series as a missing value, so every draw that contained it was recorded as a Sharpe ratio of zero. The intervals shown exclude that week; they are wider, and every one still contains zero.

**Bootstrap Implementation.** White’s Reality Check and Hansen’s SPA employ the stationary block bootstrap with expected block length  $\ell = 4$  weeks. The 12 formal candidates comprise five simplified asymmetry rules, the full rule, three momentum rules, two mean-reversion rules and always-long. Bootstrap replications: 1,000. The formal benchmark is zero weekly return.

**Candidate Universe.** The formal tests are run on the twelve real candidate strategies. A seeded random position sequence is reported alongside as a diagnostic but is excluded from the formal universe: a data-snooping correction should control the error rate across the space actually searched, and a coin-flip rule was never a strategy under consideration.

**White’s Reality Check.** Table 16 collects both tests. Following White (2000), we test the twelve-strategy universe against a zero-return benchmark. The Reality Check does not reject the null of no superior strategy ( $RC = 0.015$ ,  $p = 0.29$ ). Including the random sequence as a thirteenth candidate gives  $RC = 0.025$ ,  $p = 0.06$ . The direction runs opposite to what candidate-dilution intuition suggests: the random sequence has the highest realized sample mean in the universe, so removing it lowers the observed statistic more than it lowers the bootstrap distribution, and the  $p$ -value rises. Neither universe rejects.

**Hansen’s SPA Test.** The Superior Predictive Ability test (Hansen, 2005) agrees ( $SPA = 1.43$ ,  $p = 0.55$  on the twelve-strategy universe;  $p = 0.22$  including the random sequence): no candidate demonstrates superior predictive ability over the zero-return benchmark at conventional levels.

**Sizing.** Both test statistics are identical under the frozen-notional sizing alternative, and the  $p$ -values move by 0.003 and 0.001, because the universe maximum is attained by the always-long candidate, which does not depend on the asymmetry rule; Appendix F explains why that stability says nothing about the rule.

Table 16: Data-Snooping Correction Results

| Test                                                            | Statistic | $p$ -value | Interpretation  |
|-----------------------------------------------------------------|-----------|------------|-----------------|
| White’s Reality Check                                           | 0.015     | 0.29       | Not significant |
| Hansen’s SPA                                                    | 1.43      | 0.55       | Not significant |
| # Candidate Strategies                                          | 12        | —          | —               |
| <i>Including the random sequence as a thirteenth candidate:</i> |           |            |                 |
| White’s Reality Check                                           | 0.025     | 0.06       | Not significant |
| Hansen’s SPA                                                    | 2.06      | 0.22       | Not significant |

Note: Formal candidates: simplified asymmetry (5), full two-sided asymmetry (1), momentum (3), mean reversion (2), always-long (1). The seeded random sequence is reported as a diagnostic outside the formal universe. The formal benchmark is zero weekly return. All candidates use one execution lag. Stationary bootstrap: 1,000 replications, expected block length 4 weeks. Generated by `analysis/full_pipeline.py`; an earlier version of this table reported  $RC = 2.14$  ( $p = 0.042$ ), which could not be reproduced from any specification of the stated universe. The values published in version 3.0.0,  $RC = 0.02$  ( $p = 0.15$ ), came from the earlier implementation; the values shown are those of the corrected specification. An earlier draft of this revision reported  $p = 0.30$  (Reality Check) and  $p = 0.58$  (SPA) for the twelve-strategy universe, and  $SPA = 2.00$  with the random sequence, because the final, non-executable week entered every candidate as a zero return. The values shown exclude that week, which also changes the bootstrap draws.

**Interpretation.** The tests find no statistically superior candidate in either universe. This is a failure to reject superiority, not proof that every possible strategy has zero expected return. That a seeded coin-flip sequence would have posted the highest realized mean of the thirteen is not evidence about the coin flip. It shows how wide the dispersion of realized means is at this sample size, and therefore how little a ranking among these rules would mean.

## 5 Discussion

## 5.1 Why Asymmetry Exploitation Fails

The empirical findings reveal a pattern more thoroughgoing than “asymmetry exists statistically but fails economically”: under corrected measurement, most of the proposed asymmetry lacks dependence-robust support, and the one robust distributional finding does not establish a tradable edge. We identify five failure modes:

**Little Asymmetry to Exploit.** Under the primary construction only coverage alpha is robustly skewed, and the tail signal’s negative point estimate is fragile and sensitive to the weekly aggregation rule (Section 3.2). The trading rule instead uses fast and pricing skewness, whose full-sample intervals include zero. Its 15 in-sample episodes do not establish profitable predictability.

**Tail Evidence Is Inconclusive.** The GPD shape estimate for weekly return extremes (Section 4.8) is near zero, but its wide confidence interval includes both negative and positive values. The data do not establish Pareto-type heavy tails, nor do they prove exponential decay. With only 20 declustered maxima, tail classification is underpowered.

**Inactivity Under Training-Only Selection.** When the threshold is chosen on training data alone, the strategy opens only one out-of-sample holding episode in eight test years (Section 4.1). That count is too small for performance inference.

**The Rule Has a Momentum Signature by Construction.** Conditioning on the weeks the strategy is invested, its returns load negatively on time-series momentum ( $\beta_2 = -0.65$ ). Inference on the loading is method-sensitive (Section 4.7 defines both procedures): bias-reduced CR2 cluster-robust standard errors give an interval of  $[-1.27, -0.03]$  that excludes zero at  $p = 0.042$ , while the restricted wild cluster bootstrap gives  $p = 0.0506$ , just above the 5% line. Neither figure bears on the objection that matters. The loading is a property of the entry rules: the in-position weeks are selected by conditions defined on the same price series the momentum factor is built from, and a rule whose short leg fades medium-term strength on that series will load negatively on momentum during the weeks it is active whether or not any factor relationship exists. We report it because it characterizes what the strategy is. It is not evidence about the market, and we do not count it among the revision’s empirical findings.

**No Evidence of Generalization.** The identical rules transfer erratically to GBP/USD, SPY and GLD, and the tail signal’s negative point estimate reverses sign in every other market tested (Section 3.4). Nothing here establishes generalization, and nothing establishes its absence either: the transfers show that the rule does not transfer cleanly, not that it cannot.

Transaction costs are not the cause of the failure: the corrected gross return is already negative, and costs make it monotonically worse (Section 4.6).

## 5.2 What the Asymmetry Statistics Do Show

The corrected statistics still reveal genuine features of the signals’ distributions, just not the ones an asymmetry trader would want:

**Volatility-Expansion Asymmetry:** Coverage alpha’s robust right skew (1.75, block CI [1.18, 2.16]) is consistent with a familiar asymmetry in volatility dynamics— $\sigma$  can spike abruptly and decay more slowly (Bollerslev, 1986; Engle, 1982). Nothing in our results shows that this distributional feature predicts the direction of spot returns.

**A Negative Tail Point Estimate, Imprecisely Measured:** The signed tail signal has a negative skewness point estimate, indicating that the observed extreme negative magnitudes dominate the positive tail in this sample. That direction is the opposite of a lottery-ticket structure (Kumar, 2009; Bali et al., 2011) and closer to the crash-risk asymmetry of the carry literature (Brunnermeier et al., 2009; Burnside

et al., 2011). The dependence-robust interval  $[-3.10, 0.54]$  includes zero, so the estimate is too fragile to support a strategy.

**A Near-Symmetric Baseline:** The tests do not reject normality for fast alpha in this sample. That result is consistent with a near-symmetric normalized-return signal, but it does not rule out other predictive structure (Moskowitz et al., 2012).

### 5.3 Limitations

Several limitations warrant acknowledgment. First, 504 weekly observations and only 15 holding episodes provide limited power, especially for tail and trade-level inference. Second, the alpha definitions are fully disclosed here but are ad hoc and may not generalize. The tail signal in particular is aliased by its sampling rule: it is constructed from daily exceedances but read only on Fridays, so 70 of the 105 weeks containing an exceedance enter the panel as zeros and only 35 of 504 weekly observations are non-zero. Every tail-alpha statistic in this paper is computed on that sparse series, which is the proximate reason its skewness interval is so wide. Table 5 reports what two all-trading-days aggregations do to the tail statistics; they agree neither with each other nor with the published rule, and for the reasons given in Section 3.2 we adopt neither. The sensitivity is the reportable finding. Third, the factor exercise uses single-pair spot proxies in place of published cross-sectional FX factors; even with 55 in-position weeks, the effective sample is small, and the negative momentum loading is descriptive, not a stable structural estimate. Fourth, costs use stylized spreads and indicative mid-prices, omit financing, charge no fixed or minimum per-order component, and assume fills at the indicative first-session open; no executable bid or ask quotes are used. Fifth, cross-market transfer omits pair-specific financing and costs, asset-specific calibration of the signals, and dependence-aware inference with multiple-testing correction. Finally, Yahoo history can be revised. The original raw snapshots were not committed, so exact reproduction of the archived figures is impossible without the original raw files; the revised pipeline caches local inputs and records hashes and software versions.

**The Limits of the Verification Used Here.** This revision leans on machine checking: every free parameter is declared in one module from which the specification table is generated, restored historical footnotes were verified verbatim when they were restored, two arithmetic identities are checked by the test suite against the committed output, and the tables carrying the headline performance, factor and data-snooping results are tied cell by cell to named fields of the replication output. That last mechanism is recent and its coverage is partial: it reaches fourteen of the twenty-three tables (fifteen counting the specification table, which is generated from and checked against the specification module) and, among prose figures, the regression  $t$ -statistics only. Figures outside it are checked less formally, and we do not claim otherwise. Three of the errors corrected here were found by computing a quantity twice by different routes and requiring agreement; none was found by re-reading the code that contained it.

That apparatus has a further limitation. A test that cannot fail is indistinguishable, in its output, from a test that passes. Nine checks written during this revision passed while being structurally incapable of detecting the defect they were written for, most because they ran on data where the quantity they examined could not move. Of these, two arose while we were deliberately testing whether the existing checks could fail, which is to say while we were actively looking for that flaw. One was subtler and is the reason this paragraph exists: it was exercised on a deliberately defective input, it failed as required, and it failed for a reason incidental to the defect class it was written to detect, so that the successful detection was itself misleading evidence. And one was a guard written to prevent incorrect status figures, which produced one: it counted the test suite through a subprocess that silently resolved to an interpreter lacking the project's dependencies, collected less than half the suite, and reported the reduced figure as a

success. A check may therefore appear to succeed for a reason unrelated to the failure mode it claims to test. The remedy adopted is to require a check to fail on a deliberately defective input, and to establish that it fails for the stated reason, before its passes are credited. Several of the principal checks have been exercised this way; this is not a property we claim for every check in the repository.

Another limit is structural and does not depend on how carefully any check is built. Every mechanism described here compares a reported number against the output that produced it, so all of them are silent on a sentence whose figures are correct and whose conclusion does not follow from them. Two claims corrected during this revision were of that kind: one explained a result through a candidate universe the reported test no longer used, and one stated an inferential outcome as a realized one. Every figure in both was accurate. Numerical provenance certifies arithmetic, not reasoning, and the only instrument we found for the second is reading a sentence and asking what its numbers would have to mean for it to be true.

The converse bound deserves equal weight, because it points the other way and we met it in practice. A search that fails to establish where a figure came from is, until the search mechanism has itself been validated, evidence about the search rather than about the figure. One quantity in this paper was provisionally classified as untraceable and was within a step of being deleted; it was correct, and the search had been asking for the wrong quantity. Absence of established provenance is not evidence that a number is wrong. It licenses further tracing through version history and superseded outputs, and it licenses removal only once that tracing has itself been shown capable of succeeding.

One check is reported with a measured blind spot and left unrepaired. The identity tying the factor-regression intercept to the strategy's own mean weekly return verifies that the regression was run on the series the paper reports; it detects a substitution between distant specifications, but one nearby alternative specification remained within the pre-specified tolerance and went undetected. The identity can therefore confirm internal consistency while remaining blind to a consistently wrong underlying return series. We have not tightened the tolerance, because doing so after observing which substitution it missed would be a threshold chosen on outcomes, which is the practice this paper objects to elsewhere. The full audit record, including the mutations attempted and their magnitudes, is in the replication repository.

## 6 Conclusions

This paper investigated whether distributional asymmetries in FX alpha signals represent exploitable inefficiencies. Table 17 sets out each claim tested, the evidence for it and the verdict. The answer is negative for the economic claim, and under the primary construction the statistical premise holds for only one of five signals, which the rule does not use.

Table 17: Summary of Claims Tested, Evidence, and Verdicts

| Claim tested                                         | Evidence         | Verdict                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The alpha signals are asymmetrically distributed     | Tables 2–4       | Under the primary construction only coverage alpha has a dependence-robust skewness interval that excludes zero; the other four intervals include zero.                                                                                                                                                                                                                                    |
| The tail signal is negatively skewed                 | Table 5          | Not established. Across three weekly aggregations of the identical daily exceedance rule the estimate changes sign and the interval excludes zero under one; we report the sensitivity and do not select among the aggregations.                                                                                                                                                           |
| Weekly returns have Pareto-type heavy tails          | Tables 21 and 22 | Not established. The GPD shape estimate is too imprecise to establish heavy tails or exponential decay; clustering appears weak but is estimated from few exceedances.                                                                                                                                                                                                                     |
| The asymmetry rule earns a positive return           | Tables 6 and 7   | No. The corrected rule returns $-0.73\%$ gross over 15 holding episodes (Sharpe $+0.005$ ), with bootstrap intervals that include zero; walk-forward selection opens one episode in eight test years.                                                                                                                                                                                      |
| Costs explain the failure                            | Table 12         | No. Costs worsen the already negative gross return monotonically, and no break-even cost exists: the published 19.2-pip figure presumed a positive gross return. Omitted financing, the absence of any per-order cost component, and fills assumed at indicative opening mid-quotes remain limitations.                                                                                    |
| The economic result is robust to specification       | Tables 10 and 11 | No. Four execution timings span 7.3 percentage points with no pre-specified paired contrast distinguishable from zero; four pre-specified entry rules span 9.04 points and change sign, and the earlier claim of robustness to symmetrization is withdrawn.                                                                                                                                |
| The rule carries residual alpha or a factor exposure | Tables 13 and 14 | No residual alpha. On in-position weeks the rule loads negatively on time-series momentum ( $\beta_2 = -0.65$ ; CR2 $p = 0.042$ , wild cluster bootstrap $p = 0.0506$ , so the methods disagree at the 5% line). Those weeks are selected by entry conditions defined on the same prices as the factor, so this is a mechanical property of the rule, not a finding about factor exposure. |
| Some candidate rule beats a zero-return benchmark    | Table 16         | No statistically superior candidate; exploratory cross-market outcomes are unstable (Table 19).                                                                                                                                                                                                                                                                                            |

These null findings contribute to the literature in three ways. First, they locate the failure before the familiar gap between statistical and economic significance: most of the proposed asymmetry is not supported under dependence-robust measurement, and no rule built on the one robust asymmetry is shown to beat a zero-return benchmark after the data-snooping correction (Table 16). Null results of this kind are likely more common than publication bias reveals (Harvey, 2017). Second, they document, from the inside, how easily higher-moment “stylized facts” can be manufactured by measurement choices: this paper’s own earlier version reported pronounced positive tail skewness that turned out to describe an unsigned magnitude series, right-skewed by construction. The corrected analysis is a case study in why skewness claims should be accompanied by the sign conventions, sampling frequency, and dependence-robust intervals that make them checkable. Third, they provide a documented negative result that may spare other researchers this particular dead end.

The methodological lesson is sharper than “backtests flatter”: most of the asymmetry did not survive measurement. Distributional anomalies in financial data are easily overstated at the measurement stage, before a single trade is simulated, and the sequence of checks (signed variables, dependence-robust intervals, training-data-only selection, data-snooping correction) dismantled this one layer by layer. What remained of the economic result then changed sign with design choices the original specification never

argued for.

Future research might examine whether asymmetry exploitation is viable at higher frequencies (intraday) where both signal counts and cost structures scale differently, or in markets with different microstructure. A related study of cryptocurrency markets found a similar fragility in a different asymmetry. There, an infrastructure-versus-regulatory volatility differential that a naive test had called significant is not significant under the primary dependence-robust bootstrap, while a sensitivity analysis rejects equality one-sided, so the verdict depends on the inference scheme (Farzulla, 2026). Our results support a high burden of proof for asymmetry-based strategies: an apparently sharp effect is insufficient evidence of economic value, and is often not robust evidence of a statistical effect either.

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## Declarations

**Conflict of Interest.** The authors declare no competing interests.

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**Data Availability.** The pipeline retrieves Yahoo Finance series or reads local cached CSV files. The analysis manifest records retrieval dates, row counts, software versions, and SHA-256 hashes. Raw snapshots are not committed because redistribution terms may apply. The original snapshots underlying the previously committed results were not in the repository; exact reproduction of those results requires them.

**Code Availability.** Full replication code is available at <https://github.com/dissensus-ai/alpha-asymmetry>. The repository’s only licence file grants the Creative Commons Attribution 4.0 International licence (CC BY 4.0); no separate software licence is granted.

**AI Assistance.** Claude (Anthropic; Claude Opus and Claude Fable 5 models) assisted both authors, in distinct phases, each author directing and remaining answerable for the assistance used in their own. M. Farzulla directed its use in the original version for manuscript drafting, replication-pipeline development and code review, and statistical exposition. T. Israfilov directed its use for the corrections in the present version: reading the published replication code against the published specification, implementing the corrections, regenerating the analysis, and drafting the revised text. M. Farzulla directed its use in the October 2026 revision of this version: editing and restructuring the text, checking citations against their sources, correcting the return bootstrap’s handling of the non-executable final week, and moving the execution-timing contrasts and the weekend-gap test into the replication pipeline. ChatGPT (OpenAI; GPT-5.6 “Sol”) provided adversarial cross-review of the empirical claims. Citation verification drew on [scite](#), among other automated research tools.

Because implementing corrections is a broader role than assisting with prose, the work was structured so that it can be checked rather than trusted. Every change is a separate commit recording its rationale; the derivation of each correction from the published version is documented in the repository’s correction changelog; the figures in fourteen of the twenty-three tables (fifteen counting the generated specification table) and the regression  $t$ -statistics in the text are machine-checked against the pipeline’s own output, and the rest are checked less formally (Section 5.3); every free parameter is declared in one module from which the manuscript’s specification table is generated, and a test fails if the two disagree; the restored provenance footnotes were checked by machine for verbatim preservation when they were restored; and the analysis is covered by deterministic tests. Two arithmetic identities, the factor intercept against the strategy’s own mean weekly return and the volatility-regime returns compounding to the full-sample return, are checked by the test suite against the committed output.

That apparatus caught errors that review by reading did not, including a passage in the factor-attribution section whose stated reasoning ran opposite to its own conclusion. We regard it as part of the contribution rather than as housekeeping: it is what allows a reader to check these corrections instead of taking them on trust, and it is reusable by anyone auditing a quantitative result produced the same way. All research design, data analysis, and interpretation are solely the authors’, who take full responsibility for all claims, judgments, and errors.

**Author Contributions.** **M.F.:** conceptualisation, methodology, software, formal analysis, data curation, writing — original draft. **T.I.:** validation (audit of the published analysis code against the published specification); formal analysis (corrected estimation and small-cluster inference); software (correction of the replication pipeline, verification tooling); writing — review & editing.

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## A Enlarged Figures

For improved readability, this appendix reproduces all figures at enlarged scale.

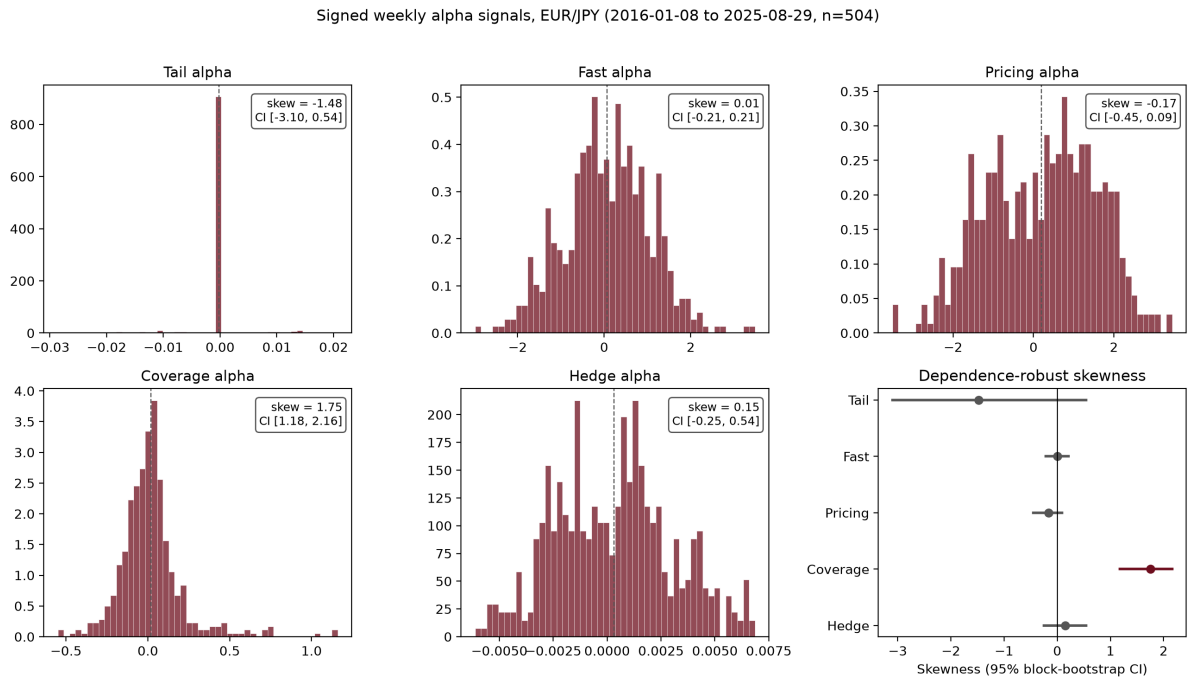


Figure 3: Alpha Asymmetry Analysis (enlarged). Distributional properties of five alpha types in the EUR/JPY analysis-ready sample (January 2016–August 2025).

Strategy equity curves and drawdowns, one-lag Friday-close convention

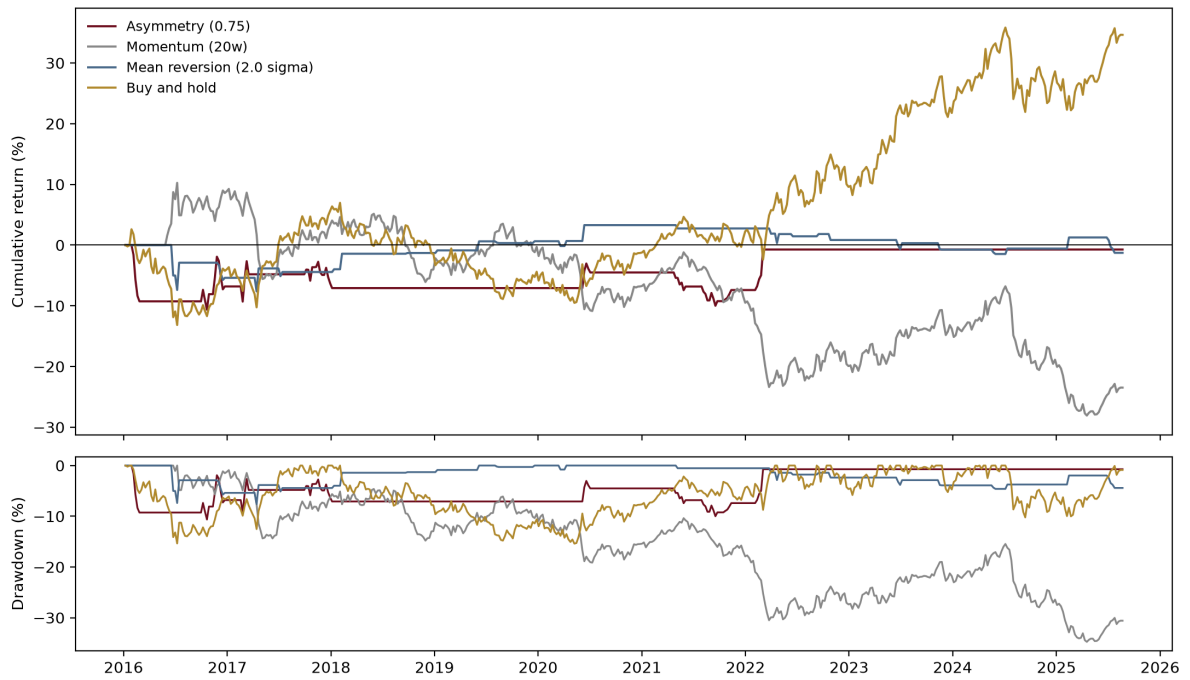


Figure 4: Strategy equity curves and drawdowns (enlarged). Cumulative gross returns and drawdown profiles for the asymmetry strategy, the 20-week momentum and  $2.0\sigma$  mean-reversion benchmarks, and buy-and-hold on the 504-week EUR/JPY sample.

## B Standard Estimators and Test Statistics

This appendix gives the standard estimators and normality tests described in Sections 2.3 and 2.4 and reported in Tables 2, 3 and 18.

**Normality Tests.** We assess departures from normality with three standard tests, each with distinct power properties against different alternatives. The Shapiro-Wilk test (Shapiro and Wilk, 1965) has high power against asymmetric and heavy-tailed alternatives for moderate sample sizes. The D’Agostino-Pearson omnibus  $K^2$  test (D’Agostino et al., 1990) combines normalizing transformations of sample skewness and kurtosis and is particularly powerful against asymmetric alternatives. As a robustness check, the Jarque-Bera test (Jarque and Bera, 1980) is a Lagrange multiplier test based on the same two moments.

**Normality Test Results.** Four of five alpha types reject normality. For fast alpha, none of the reported tests rejects the Gaussian null (SW  $p = 0.55$ , JB statistic  $\approx 0$ ,  $K^2$   $p = 0.99$ ). This is compatibility with the null in this sample, not proof of exact Gaussianity. Tail and coverage depart severely; pricing and hedge reject more mildly. All three normality tests assume independent observations, which the overlapping rolling constructions violate, so these rejections are indicative; dependence-robust versions exist (Bai and Ng, 2005) but are not computed here.

Table 18: Normality Tests

| Alpha    | SW       | JB      | $K^2$    |
|----------|----------|---------|----------|
| Tail     | 0.335*** | 7922*** | 244.4*** |
| Fast     | 0.997    | 0.0     | 0.0      |
| Pricing  | 0.984*** | 13.3**  | 30.1***  |
| Coverage | 0.865*** | 1489*** | 217.6*** |
| Hedge    | 0.990**  | 7.4*    | 11.7**   |

Note: SW = Shapiro-Wilk statistic; JB = Jarque-Bera statistic;  $K^2$  = D’Agostino-Pearson omnibus statistic; signed weekly series,  $n = 504$ . \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$  (unadjusted). All three tests assume independent observations. Generated by analysis/full\_pipeline.py.

**Sample Skewness.** For a sample  $\{x_1, \dots, x_n\}$  with sample mean  $\bar{x}$  and standard deviation  $s$ , we compute the adjusted Fisher-Pearson coefficient:

$$\hat{\gamma}_1 = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left( \frac{x_i - \bar{x}}{s} \right)^3 \quad (\text{B.1})$$

This bias-corrected estimator is consistent under standard regularity conditions. The exact finite-sample standard error of  $\hat{\gamma}_1$  under normality,  $\sqrt{6(n-2)/[(n+1)(n+3)]}$ , differs from the approximation  $\sqrt{6/n}$  used in Table 3 only in the third decimal at  $n = 504$ .

**Excess Kurtosis.**

$$\hat{\gamma}_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^n \left( \frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)} \quad (\text{B.2})$$

This estimator subtracts 3 (the kurtosis of a normal distribution) and applies finite-sample bias correction.

**Shapiro-Wilk Test.** The Shapiro-Wilk statistic (Shapiro and Wilk, 1965) tests the null hypothesis

$H_0 : X \sim \mathcal{N}(\mu, \sigma^2)$  against general alternatives:

$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (\text{B.3})$$

where  $x_{(i)}$  are order statistics and  $a_i$  are tabulated coefficients.

**D’Agostino-Pearson Omnibus Test.** Following D’Agostino et al. (1990), skewness and kurtosis are tested jointly using the  $K^2$  statistic:

$$K^2 = Z_1(\hat{\gamma}_1)^2 + Z_2(\hat{\gamma}_2)^2 \sim \chi^2_2 \quad (\text{B.4})$$

where  $Z_1$  and  $Z_2$  are normalizing transformations of sample skewness and kurtosis.

**Jarque-Bera Test.** The Jarque-Bera statistic (Jarque and Bera, 1980) is

$$JB = \frac{n}{6} \left( \hat{\gamma}_1^2 + \frac{(\hat{\gamma}_2)^2}{4} \right) \stackrel{a}{\sim} \chi^2_2 \quad (\text{B.5})$$

This Lagrange multiplier test is asymptotically equivalent to the likelihood ratio test for normality.

## C Cross-Market Check

For cross-market validation, we apply the same transformations to GBP/USD, SPY (S&P 500 ETF), and GLD (Gold ETF) daily closes from Yahoo Finance, constructing tail, fast, pricing, and coverage alphas identically to EUR/JPY. Hedge alpha is excluded from the cross-market set because the DXY-correlation proxy was specified for the EUR/JPY setting and is not a genuine transferable rate-differential signal.

To test whether the EUR/JPY findings generalize, we apply the identical alpha construction and the identical trading strategy (unchanged code, unchanged parameters) to GBP/USD, SPY, and GLD (Section 2.1). Hedge alpha is excluded as EUR/JPY-specific; the four remaining signals and the two-sided asymmetry strategy (0.75 threshold) transfer directly. Table 19 reports the signed weekly skewness of each signal alongside the strategy’s gross cumulative return and, for scale, each market’s buy-and-hold return over the same 504-week window.

Table 19: Cross-Market Check

| Market  | Tail Skew | Fast Skew | Pricing Skew | Coverage Skew | Strat. Ret. | B&H Ret. |
|---------|-----------|-----------|--------------|---------------|-------------|----------|
| EUR/JPY | −1.48     | 0.01      | −0.17        | 1.75          | −0.73%      | 33.53%   |
| GBP/USD | 0.82      | 0.01      | −0.14        | 1.58          | −14.45%     | −7.63%   |
| SPY     | 0.94      | −0.21     | −1.09        | 1.85          | 14.71%      | 294.30%  |
| GLD     | 0.69      | −0.00     | −0.19        | 1.32          | −13.03%     | 200.97%  |

Note: Bias-corrected skewness of signed weekly signals ( $n = 504$  per market). Strategy returns use the corrected one-lag strategy with 15, 19, 4, and 17 directional holding episodes respectively. SPY uses adjusted prices, explicitly requested from the data provider. Point estimates do not include market-specific block-bootstrap inference or costs; this remains exploratory. Generated by `analysis/full_pipeline.py`; an earlier version of this table reported aggregated “MR/TF/HAT” skewness categories and returns that traced to no committed code, and has been replaced.

The distributional signature does not travel: EUR/JPY’s negative tail-skew point estimate is positive in all three other markets, while coverage alpha’s right skew recurs. Strategy outcomes are likewise unstable: the unchanged rules lose 14.45% on GBP/USD, gain 14.71% on SPY while trailing buy-and-hold by roughly 280 percentage points, and lose 13.03% on GLD. These exploratory transfers lack pair-specific costs and dependence-aware inference, so they do not establish effects in any market. They

are an illustrative transfer check, not a multi-asset analysis, which would need asset-specific signal calibration, market-specific cost models and asset-by-asset inference with multiple-testing correction (Section 5.3).

## D Paired Execution-Timing Contrasts

Table 20 reports the six pairwise contrasts among the four execution timings of Section 4.4. The three primary contrasts were specified before the Monday-close and Tuesday-open results were computed; the other three are secondary.

Table 20: Paired Execution-Timing Contrasts Under Four Bootstrap Schemes

| Contrast (later – earlier)  | Role      | Estimate | 4-week block  | 8-week block  | 13-week block | Stationary    |
|-----------------------------|-----------|----------|---------------|---------------|---------------|---------------|
| Monday open – Friday close  | Primary   | 0.63     | [−0.67, 2.03] | [−0.36, 2.07] | [−0.13, 1.99] | [−0.60, 2.00] |
| Monday close – Monday open  | Primary   | −0.02    | [−0.08, 0.04] | [−0.08, 0.04] | [−0.07, 0.04] | [−0.08, 0.04] |
| Tuesday open – Monday close | Primary   | −0.77    | [−2.36, 0.45] | [−2.03, 0.36] | [−1.90, 0.25] | [−2.27, 0.50] |
| Monday close – Friday close | Secondary | 0.61     | [−1.03, 2.26] | [−0.86, 2.08] | [−0.73, 1.95] | [−0.93, 2.16] |
| Tuesday open – Friday close | Secondary | −0.15    | [−2.21, 1.90] | [−1.93, 1.62] | [−1.91, 1.60] | [−2.23, 1.92] |
| Tuesday open – Monday open  | Secondary | −0.79    | [−2.47, 0.90] | [−2.27, 0.70] | [−2.14, 0.57] | [−2.43, 0.85] |

Note: Difference in geometric annualized gross return, in percentage points, on the common sample of 502 weeks (Table 10). Each replicate resamples one set of weeks and applies it to all four timings; the position path is held fixed. Moving-block schemes draw overlapping blocks of fixed length; the stationary bootstrap uses an expected block of four weeks;  $B = 2000$  throughout. Primary rows: 95% percentile intervals. Secondary rows: 95% simultaneous intervals for the three secondary contrasts, from the maximum standardized deviation across them. Generated by `analysis/full_pipeline.py` (`execution_timing.paired_contrasts`).

## E Tail Distribution of Weekly Returns

A concern with any exceedance-based signal is whether extreme observations reflect a stable tail distribution or sparse outliers that may not recur out-of-sample. As a separate diagnostic, we characterize absolute Friday-close-to-Friday-close EUR/JPY returns using Extreme Value Theory (McNeil and Frey, 2000; Coles, 2001). This is not the same sample as tail alpha, which flags daily exceedances before Friday sampling. The exercise therefore describes weekly return magnitudes and does not establish an upside-asymmetry story.

**Methodology.** We fit a Generalized Pareto Distribution (GPD) to cluster maxima above the 95th percentile threshold. The GPD is characterized by shape parameter  $\xi$  (determining tail heaviness) and scale parameter  $\sigma$ . Positive  $\xi$  indicates Pareto-type heavy tails;  $\xi = 0$  corresponds to exponential decay; negative  $\xi$  implies bounded tails.

**Exceedance Preprocessing.** Financial tail events exhibit temporal clustering (volatility clustering), violating the GPD independence assumption. We apply runs declustering with minimum inter-exceedance separation of 5 weeks, following Coles (2001). Of the 26 raw exceedances above the 95th percentile threshold, declustering yields 20 cluster maxima used for GPD estimation.

A five-week separation is aggressive for weekly data, where persistence is far shorter than in the daily series runs declustering was developed for. We therefore report what a shorter window does. At a one-week separation, the shortest declustering that does anything at all, the fit uses 22 cluster maxima instead of 20 and gives  $\hat{\xi} = -0.209$  against  $-0.248$ , with a confidence interval of  $[-1.53, 0.22]$  against  $[-1.51, 0.25]$ . Two weeks gives 21 maxima and  $-0.241$ . The full grid is in Table 21.

Table 21: Sensitivity of the GPD Fit to the Declustering Separation

| Separation         | Cluster maxima | $\hat{\xi}$ | 95% CI        | KS $p$ |
|--------------------|----------------|-------------|---------------|--------|
| 1 week             | 22             | −0.209      | [−1.53, 0.22] | 0.997  |
| 2 weeks            | 21             | −0.241      | [−1.51, 0.22] | 0.981  |
| 3 weeks            | 20             | −0.248      | [−1.51, 0.25] | 0.971  |
| 5 weeks (reported) | 20             | −0.248      | [−1.51, 0.25] | 0.971  |

Note: GPD fitted by maximum likelihood to cluster maxima above the 95th percentile of weekly absolute returns.  $\hat{\xi}$  CI from a nonparametric bootstrap of cluster maxima,  $B = 1000$ . The 5-week row therefore differs slightly from Table 22, which uses  $B = 2000$ .

The shortest separation therefore recovers 2 additional cluster maxima and moves the shape estimate by 0.04, inside intervals roughly 1.75 wide. The imprecision of the tail estimate is a consequence of having 26 exceedances in ten years, not of how they are declustered, and no conclusion in this section changes at any separation we tried. We estimate the extremal index  $\theta = 0.83$  via the Ferro–Segers intervals estimator (Ferro and Segers, 2003), with a 95% confidence interval of  $[0.60, 1.00]$  obtained by bootstrap resampling of the inter-exceedance times themselves ( $B = 2000$ ). That scheme preserves the gap distribution the estimator is built on and, unlike a naive index bootstrap, produces an interval containing its own point estimate.

Table 22: GPD Parameter Estimates: Weekly Absolute Returns (Declustered)

| Parameter                   | Estimate | 95% CI         |
|-----------------------------|----------|----------------|
| Shape ( $\xi$ )             | −0.25    | [−1.49, 0.27]  |
| Scale ( $\sigma$ )          | 0.0089   | —              |
| Threshold (95th pctl)       | 0.0244   | —              |
| Extremal index ( $\theta$ ) | 0.83     | [0.60, 1.00]   |
| Raw exceedances             | 26       | —              |
| Cluster maxima              | 20       | —              |
| KS test $p$ -value          | 0.971    | Fail to reject |

Note: GPD fitted via maximum likelihood to cluster maxima of weekly absolute returns above the 95th percentile after runs declustering (5-week separation).  $\xi$  CI from a nonparametric bootstrap of cluster maxima ( $B = 2000$ );  $\theta$  CI from bootstrap resampling of inter-exceedance times ( $B = 2000$ ). KS = Kolmogorov-Smirnov goodness-of-fit test. Generated by `analysis/full_pipeline.py`; an earlier version of this table reported a  $\theta$  confidence interval of  $[0.42, 0.63]$  that excluded its own point estimate—an invalid interval produced by a biased index-resampling scheme, replaced here.

**Tail Characterization.** Table 22 reports the fit. The estimated shape parameter  $\hat{\xi} = -0.25$  (95% CI:  $[-1.49, 0.27]$ ) is imprecise. The interval admits bounded, exponential-type, and moderately Pareto-type behavior; the data therefore do not support a strong tail-classification claim. The Kolmogorov-Smirnov test does not reject the fitted GPD ( $p = 0.97$ ), but with only 20 cluster maxima it has limited diagnostic power.

**Clustering Dynamics.** The extremal index  $\theta = 0.83$  indicates weak temporal clustering in tail events: the mean cluster size is  $1/\theta \approx 1.2$ , and the confidence interval reaches 1.0 (no clustering). Exceedances arrive close to independently, with mild GARCH-type persistence (Bollerslev, 1986) at most. Clustering therefore does little to reduce the effective sample size for tail inference; the binding constraint is simply the small number of exceedances.

**Implications.** EVT provides no affirmative evidence for the lottery-like right-tail mechanism required by the trading story. It does not justify assuming a particular tail class or finite variance outside

this sample.

## F Position-Sizing Alternative

Section 2.6 adopts weekly re-evaluation of the asymmetry index while a direction is held. Table 23 compares it with freezing the notional at entry, the other extension of the published rule to held-but-unsigalled weeks.

Table 23: Position Sizing: Weekly Re-evaluation and Notional Frozen at Entry

| Sizing                          | Return | Sharpe | MDD     | Episodes | Legs | Resizes | Turnover |
|---------------------------------|--------|--------|---------|----------|------|---------|----------|
| Weekly re-evaluation (reported) | −0.73% | 0.005  | −10.64% | 15       | 61   | 31      | 52.00    |
| Frozen at entry                 | −2.01% | −0.023 | −12.17% | 15       | 30   | 0       | 49.15    |

Note: Cumulative gross return over the 503 weeks with a realizable return; both rules hold the same 55 weeks in the same 15 episodes and differ only in whether the notional follows the weekly asymmetry index. Turnover in gross-notional units. Generated by `analysis/full_pipeline.py (sizing_variants)`.

**Why These Statistics Are Insensitive to the Sizing Specification.** Both test statistics are numerically identical under the weekly and frozen-notional sizing specifications, and the  $p$ -values differ by 0.003 and 0.001 respectively. This is not a transcription error, and the reason is a property of the formal universe that we verify in the replication output. Of the twelve formal candidates, only the full asymmetry strategy depends on the sizing convention at all; the other eleven are built with fixed unit sizing or are buy-and-hold. Both tests take a maximum across that universe, and the maximum is attained by the always-long candidate, whose returns do not depend on the asymmetry rule in any respect. The observed statistic therefore cannot move when the sizing changes. The  $p$ -value can, because the bootstrap distribution is computed across the whole universe and does include the one candidate that changed, which is why the shift is small but not zero. A reader should expect these two rows to be stable across any change confined to the asymmetry rule, and should not read that stability as evidence about the rule. The argmax, the identity of the two statistics and the two  $p$ -value differences are all recorded in `data_snooping.sizing_invariance` in the replication output.